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Simulation of partial melting material

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Simulation of partial melting material

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Dedication

To all future Longhorn graduate students.

Epigraph

What starts here changes the world.

—University of Texas at Austin

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Thanks to the previous authors of this template, and to the Graduate School for providing it.

Preface

Readers, this document represents a great deal of time and effort. Enjoy. Cheers, Chenyu Tian.

Abstract

Simulation of partial melting material

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Chapter 1: Governing Equations in Mantle Dynamics

1.1 Mechanics in Mantle Dynamics

Extensive studies have been conducted on fluid dynamics of a two-phase mixture. We briefly discuss the four governing equations of mechanics, drawing on a similar discussion presented by Drew and Segel (1971), Mckenzie (1984) and Katz (2022).

1.1.1 Phase Fraction

We begin with defining the phase fraction using an indicator function, under the assumption that multiple phases cannot coexist at the same spatial location. Accordingly, the phases are considered immiscible. The indicator function is defined as

$$\chi_i(\mathbf{x}) = \begin{cases} 1 & \text{phase } i, \\ 0 & \text{not phase } i. \end{cases} \tag{1.1}$$

We then define the macroscopic phase fraction function of phase i within a representative volume element V as

$$\phi_i = \frac{1}{V} \int_V \chi_i(\mathbf{x}) d\mathbf{v}, \tag{1.2}$$

where $\sum_i \phi_i = 1$. In the incompressible setting, the phase fraction may be equivalently regarded as the volume fraction occupied by each phase within a representative elementary volume, reflecting the proportion of space occupied by each phase at the macroscopic scale. Phase-averaged (or volume-averaged) quantities, also referred to as bulk quantities, are defined as

$$\overline{\gamma} = \sum_i \phi_i \gamma_i, \tag{1.3}$$

where γ_i denotes a quantity that may be scalar- or vector-valued.

1.1.2 Conservation Laws

We consider the conservation equation for a single-phase system. For a given volumetric quantity γ defined within a representative volume element V, we can write a corresponding conservation equation as:

$$\frac{d}{dt}\int_V \gamma d\mathbf{v} = -\int_{\partial V} \mathbf{f}\cdot d\sigma + \int_V S d\mathbf{v}, \tag{1.4}$$

where $\mathbf{f}$ represents flux travelling across the boundary of a representative volume element V. The minus sign indicates that the positive direction aligns with normal unit vector pointing outwards. The source/sink term S, represents the volumetric production rate of the volumetric quantity γ. We apply Gauss's theorem to obtain the differential form

$$\frac{\partial \gamma}{\partial t} + \nabla\cdot\mathbf{f} = S. \tag{1.5}$$

We model the partially molten mantle as a two-phase mixture consisting of liquid magma and solid mantle. The phase fraction of liquid magma is denoted by ϕ_l and phase fraction of solid mantle is labelled as ϕ_s, such that the relation $\phi_l + \phi_s = 1$ is always obtained.

We rewrite the single-phase conservation law (1.5) with the bulk density as

$$\frac{\partial \overline{\rho}}{\partial t} + \nabla\cdot\overline{\rho\mathbf{v}} = 0, \tag{1.6}$$

where $\overline{\rho} = \phi_l\rho_l + \phi_s\rho_s$. Under the Boussinesq approximation, we neglect the compressibility of each individual phase. Furthermore, we extend the Boussinesq approximation by disregarding the density difference between the two phases, except in the body force term. Accordingly, the continuity equation for the two-phase mixture can be written as

$$\nabla\cdot\overline{\mathbf{v}} = 0. \tag{1.7}$$

Similarly, the conservation equation for the phase-averaged momentum takes the form

$$\frac{\partial\overline{\rho\mathbf{v}}}{\partial t} + \nabla\cdot\overline{\rho\mathbf{v}\otimes\mathbf{v}} = \overline{\rho}\mathbf{g} + \nabla\cdot\overline{\boldsymbol{\tau}} - \nabla\overline{p}, \tag{1.8}$$

where $\boldsymbol{\tau}$ represents the deviatoric stress. By neglecting inertial effects, justified by the low Reynolds number of magma and mantle, the static force balance equation reduces to

$$\overline{\rho}\mathbf{g} + \nabla \cdot \overline{\boldsymbol{\sigma}} = 0, \tag{1.9}$$

where $\boldsymbol{\sigma} = \boldsymbol{\tau} - p\boldsymbol{I}$ represents the total stress, consisting of the deviatoric component $\boldsymbol{\tau}$ and the isotropic pressure $p\boldsymbol{I}$.

1.1.3 Darcy's Law

The equations governing bulk quantities (1.7) and (1.9) alone are insufficient to simultaneously determine the two distinct velocities and pressures. Therefore, an additional constitutive relationship is required to describe the behavior of the liquid phase within the porous medium. We examine more closely the interphase forces between the solid and liquid phases. A brief overview of the derivation of the individual governing equations for each phase is provided here, while more detailed and rigorous treatments can be found in the mixture theory literature, such as Bowen (1980). We write momentum balances for each individual phase as:

$$\begin{aligned} 0 &= \phi_l \rho_l \mathbf{g} + \nabla \cdot (\phi_l \boldsymbol{\tau}_l) - \nabla(\phi_l p_l) + \mathbf{F}_l, \\ 0 &= \phi_s \rho_s \mathbf{g} + \nabla \cdot (\phi_s \boldsymbol{\tau}_s) - \nabla(\phi_s p_s) - \mathbf{F}_s, \end{aligned} \tag{1.10}$$

where $\mathbf{F}_l = \mathbf{F}_s$ are the interphase forces between solid and liquid phases. The interphase forces between the solid and liquid phases are equal and opposite, in accordance with Newton's third law. Neglecting the deviatoric stress in the liquid phase and retaining only the isotropic pressure component, we balance the remaining stress with the viscous drag force and the equilibrium body force. Accordingly, we obtain the following relation for the relative velocity between phases, in alignment with Darcy's law

$$\phi_l(\mathbf{v}_l - \mathbf{v}_s) + \frac{k_{\phi_l}}{\mu_l}(\nabla p_l - \rho_f \mathbf{g}) = 0, \tag{1.11}$$

where k_{ϕ_l} is the matrix permeability, which usually takes the quadratic form as $k_{\phi_l} = k_0 \phi_l^2$.

1.1.4 Compaction

In the context of mantle dynamics, an additional yet unique and essential relationship regulates melt migration. We now revisit the continuous equation (1.6), and consider the two phases separately with the effects of melting incorporated, i.e. ,

$$\begin{aligned} \frac{\partial \phi_l}{\partial t} + \nabla \cdot (\phi_l \mathbf{v}_l) = \Gamma, \\ \frac{\partial (1-\phi_l)}{\partial t} + \nabla \cdot ((1-\phi_l)\mathbf{v}_s) = -\Gamma, \end{aligned} \tag{1.12}$$

where Γ is the unknown melting rate. We focus on the solid phase equation to get the evolving compaction relationship as

$$\frac{\partial \phi_l}{\partial t} + \mathbf{v}_s \cdot \nabla \phi_l = \Gamma + (1-\phi_l)\nabla \cdot \mathbf{v}_s, \tag{1.13}$$

where the divergence term $\nabla \cdot \mathbf{v}_s$ is often referred to as the compaction rate. We approximate the static compaction relation as

$$\xi_{\phi_l} \nabla \cdot \mathbf{v}_s = p_l - p_s, \tag{1.14}$$

where ξ_{ϕ_l} is often referred to as bulk viscosity or compaction viscosity. The form of the compaction viscosity ξ_{ϕ_l} has been the subject of extensive discussion. Mckenzie (1984) popularized the concept of compaction viscosity and relating the term to the stress balance in a partially molten system. Bercovici et al. (2001) extended this formulation by incorporating surface tension effects. Rabinowicz et al. (2002) further examined compaction behavior in the limit of infinitesimally small porosity. Following the approaches of Hewitt and Fowler (2008) and Katz (2008), we approximate the compaction viscosity ξ_{ϕ_l} using a simplified constitutive relationship,

$$\xi_{\phi_l} = \frac{\mu_s}{\phi_l}, \tag{1.15}$$

which highlights the fact that increased melting facilitates the compaction of the solid phase, while in the limit of no melting ($\phi_l = 0$), compaction is entirely eliminated.

1.1.5 Full Static System

By combining equations (1.7), (1.9), (1.11) and (1.14), We finally arive at the full static flow mechanical system,

$$\begin{aligned}
\phi_l(\mathbf{v}_l - \mathbf{v}_s) + \frac{k_{\phi_l}}{\mu_l}(\nabla p_l - \rho_f \mathbf{g}) &= 0, && (1.16)\\
\nabla \cdot (\phi_l \mathbf{v}_l + \phi_s \mathbf{v}_s) &= 0, && (1.17)\\
\nabla \bar{p} - \nabla \cdot \boldsymbol{\tau}(\mathbf{v}_s) + \bar{\rho}\mathbf{g} &= 0, && (1.18)\\
\mu_s \nabla \cdot \mathbf{v}_s - \phi_l(p_l - p_s) &= 0, && (1.19)
\end{aligned}$$

where the deviatoric stress of the solid is

$$\boldsymbol{\tau}(\mathbf{v}_s) = 2\mu_s \phi_s \big(\mathcal{D}\mathbf{v}_s - \tfrac{1}{3}\nabla \cdot \mathbf{v}_s \mathbf{I}\big), \tag{1.20}$$

and $\mathcal{D}\mathbf{v}_s = \frac{1}{2}(\nabla \mathbf{v}_s + \nabla \mathbf{v}_s^T)$ is the symmetric gradient. Moreover, we adjust the relative permeability $k_{\phi_l} = k_0 \phi_l^{2+2\Theta}$ with constant $\Theta \in [0, 1/2]$. Here μ_s and μ_l are dynamic viscosities for solid and liquid respectively. This complete static mechanical system relates the velocities and pressures of the solid and liquid phases to a given porosity distribution, thereby laying the foundation for subsequent computations.

1.1.6 Scaled Formulation

The Darcy equation mathematically degenerates in the regions where the porosity is zero. To handle this issue, we use the formulation developed previously by Arbogast et al. (2017). We first define pressure potentials as

$$q_l = p_l - \rho_l g z \quad \text{and} \quad q_s = p_s - \rho_l g z, \tag{1.21}$$

where $g = |\mathbf{g}|$ is the magnitude of gravity acceleration and z is the depth, such that $\mathbf{g} = g\nabla z$. Note that ρ_l is used to define both potentials, so that with $q = \bar{q} = \bar{p} - \rho_l g z$,

$$p_l - p_s = q_l - q_s = \frac{1}{1-\phi_l}(q_l - q).$$

In the mechanical system, the subscript on ϕ_l will be omitted. and we will simply write it as ϕ. Accordingly, the solid phase fraction will be expressed as $1-\phi$, emphasizing the constraint that the two phase fractions sum to one. Thus

$$\phi \mathbf{v}_r + \frac{k_0 \phi^{2+2\Theta}}{\mu_l} \nabla q_l = 0, \tag{1.22}$$

$$\mu_s \nabla \cdot (\phi \mathbf{v}_r) + \frac{\phi}{1-\phi}(q_l - q) = 0, \tag{1.23}$$

$$\nabla q - \nabla \cdot \tau(\mathbf{v}_s) = -(1-\phi)\rho_r \mathbf{g}, \tag{1.24}$$

$$\mu_s \nabla \cdot \mathbf{v}_s - \frac{\phi}{1-\phi}(q_l - q) = 0, \tag{1.25}$$

where the relative velocity $\mathbf{v}_r = \mathbf{v}_l - \mathbf{v}_s$ and the density difference $\rho_r = \rho_s - \rho_l$. Scaled variables are also defined,

$$\tilde{\mathbf{v}}_r = \phi^{-\Theta} \mathbf{v}_r \quad \text{and} \quad \tilde{q}_l = \phi^{1/2} q_l, \tag{1.26}$$

and the problem is reformulated as

$$\tilde{\mathbf{v}}_r + \frac{k_0 \phi^{1+\Theta}}{\mu_l} \nabla(\phi^{-1/2} \tilde{q}_l) = 0, \tag{1.27}$$

$$\mu_s \phi^{-1/2} \nabla \cdot (\phi^{1+\Theta} \tilde{\mathbf{v}}_r) + \frac{1}{1-\phi}(\tilde{q}_l - \phi^{1/2} q) = 0, \tag{1.28}$$

$$\nabla q - \nabla \cdot \tau(\mathbf{v}_s) = -(1-\phi)\rho_r \mathbf{g}, \tag{1.29}$$

$$\mu_s \nabla \cdot \mathbf{v}_s - \frac{\phi^{1/2}}{1-\phi}(\tilde{q}_l - \phi^{1/2} q) = 0, \tag{1.30}$$

where the new scaled relative velocity maintains numerical stability in the zero prosority regions.

1.2 Transport of Chemical Species and Energy

With the velocities of solid and liquid phases determined, we are able to evolve our system forward in time. In our system, chemical species and energy are transported by the motion of the solid and the liquid phases.

1.2.1 Transport of Chemical Species

The realistic chemical compositions of mantle and magma are highly complicated and very hard to trace at the same time. Therefore, we present a simple two-species two-phase model. In this framework, we also negelect detailed description of the chemical reaction process and exclude the effect of the change of Gibbs free energies. Similar discussion can be found in the literature like Spiegelman and Elliott (1993), Hewitt and Fowler (2008), Katz (2008) and Hesse et al. (2022). Both species can exist in either the solid or liquid phase. They are assumed to be immiscible in the solid state but fully miscible in the liquid phase. By again extending the Boussinesq approximation, we consider a homogenous density ρ for both the solid phase and the liquid phase. Figure 1.1 illustrates the coexistence of species and phases in representative volume element. Within the representative volume element V, $\gamma_{i,j}$ represents

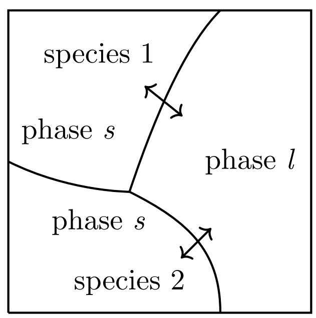


Figure 1.1: Phases and species in an representative volume element (RVE).

one quantity of the species i present in phase j, where $i \in 1, 2$ and $j \in \{s, l\}$. The mass fraction $c_{i,j}$ of a given species i in phase j is also regarded as a concentration. We can write transport equations for an individual species i, assuming a pseudo melting term Γ representing the transition between the solid and the liquid phase as

$$\frac{\partial}{\partial t}\big(\phi_l \rho_l c_{i,l}\big) + \nabla \cdot \big(\phi_l \rho_l c_{i,l} \mathbf{v}_l - \phi_l \rho_l \mathcal{D}_{i,l} \nabla c_{i,l}\big) = \Gamma, \tag{1.31}$$

$$\frac{\partial}{\partial t}\big(\phi_s \rho_s c_{i,s}\big) + \nabla \cdot \big(\phi_s \rho_l c_{i,s} \mathbf{v}_s - \phi_s \rho_s \mathcal{D}_{i,s} \nabla c_{i,s}\big) = -\Gamma, \tag{1.32}$$

where $\mathcal{D}_{i,l}$ and $\mathcal{D}_{i,s}$ are the diffusion coefficients of component i in the liquid and the solid phase respectively. The diffusion term will be dropped in the solid phase. By summing up equations (1.31) and (1.32), we obtain an advection-diffusion equation as

$$\frac{\partial \overline{\rho c_i}}{\partial t} + \nabla \cdot \left(\overline{\rho c_i \mathbf{v}} - \phi_l \rho_l \mathcal{D}_{i,l} \nabla c_{i,l}\right) = 0, \tag{1.33}$$

where the phase averaged quantities are $\overline{\rho c_i} = \sum_{j\in\{s,l\}} \phi_j \rho_j c_{i,j}$ and $\overline{\rho c_i \mathbf{v}} = \sum_{j\in\{s,l\}} \phi_j \rho_j c_{i,j} \mathbf{v}_j$. The subscript i will be omitted, as only one chemical species needs to be explicitly computed in this two-species system. By extending the Boussinesq approximation such that $\rho_l = \rho_s = \overline{\rho}$, the species transport equation can be simplified to the form

$$\frac{\partial \overline{c}}{\partial t} + \nabla \cdot \left(\overline{c\mathbf{v}} - \phi_l \mathcal{D}_l \nabla c_l\right) = 0. \tag{1.34}$$

1.2.2 Effective Velocity

The concept of effective velocity has long been employed in chemistry, particularly in the context of particle transport. In mantle dynamics, Spiegelman (1996) applied effective velocity for equilibrium transport of concentration within the liquid phases. We define a corresponding effective velocity for the transport of bulk concentration. To begin, we define the partition coefficient as $D_i = c_{i,s}/c_{i,f}$. The concentration of species i in the liquid and solid phases can be expressed in terms of the bulk concentration as

$$(\phi_l + \phi_s D_i) c_{i,l} = \overline{c}_i, \tag{1.35}$$

$$(\phi_l / D_i + \phi_s) c_{i,s} = \overline{c}_i. \tag{1.36}$$

By expressing $c_{i,l}$ and $c_{i,s}$ with bulk concentration given by equations (1.35) and (1.36), we can rewrite the bulk velocity as

$$\overline{c_i \mathbf{v}} = \phi_s \mathbf{v}_s c_{i,s} + \phi_l \mathbf{v}_l c_{i,l} = \left(\frac{\phi_l \mathbf{v}_l + D\phi_s \mathbf{v}_s}{\phi_l + \phi_s D}\right)\overline{c}_i. \tag{1.37}$$

We get the proper expression for the effective velocity, denoted as

$$\mathbf{v}_{\text{eff}} = \frac{\phi_l \mathbf{v}_l + D_i \phi_s \mathbf{v}_s}{\phi_l + \phi_s D_i}, \tag{1.38}$$

which is the appropriate velocity at which the bulk concentration should advect. In the absence of melting, where $\phi_l = 0$, we retrieve $\mathbf{v}_{\text{eff}} = \mathbf{v}_s$. Conversely, in the case of complete melting, where $\phi_l = 1$, we obtain $\mathbf{v}_{\text{eff}} = \mathbf{v}_l$. Therefore, the definition of effective velocity is well defined in those two limiting scenarios. We rewrite the transport equation (1.34) with effective velocity (1.38) as

$$\frac{\partial C}{\partial t} = \nabla \cdot (\mathbf{v}_{\text{eff}} C - \phi_l \mathcal{D}_l \nabla c_l) = 0, \tag{1.39}$$

where $C = \overline{c}$ for simplicity.

1.2.3 Transport of Energy

According to the first law of thermodynamics, we write the energy balance equation for the specfic internal energy u_i in the phase-averaged form as

$$\frac{\partial \overline{\rho u}}{\partial t} + \nabla \cdot \overline{\rho u \mathbf{v}} = \overline{\rho \mathcal{Q}} + \nabla \cdot \overline{K} \nabla T + \nabla \cdot \overline{\boldsymbol{\sigma} \cdot \mathbf{v}} + \overline{\rho \mathbf{v}} \cdot \mathbf{g}, \tag{1.40}$$

where $\overline{K} = \phi_l K_l + \phi_s K_s$ is the phase averaged thermal conductivity. On the left-hand side, the equation represents the time-dependent rate of change of internal energy of a two-phase mixture as it moves with the flow. On the right-hand side, it includes the heat source term $\mathcal{Q}$, conductive heat transfer following Fourier's law, mechanical work done by mixture stress, and work against gravity, respectively.

In a thermochemical equilibrium scenario, we replace internal energy u_i with enthalpy h_i, and use hydrostatic P as a uniform pressure, i.e., $P = \rho_l g(z - z_0)$. The specific enthalpy is defined as

$$h_i = u_i + p_i \tilde{V}_i = u_i + P/\rho_i, \tag{1.41}$$

where $\tilde{V}_i = 1/\rho_i$ is the specific volume. We formulate the enthalpy equation by neglecting the external heating/cooling term $\mathcal{Q}$, mechanical work done by mixture stress and work against gravity, following the discussion presented by Katz (2008), so

$$\frac{\partial \overline{\rho h}}{\partial t} + \nabla \cdot \overline{\rho h \mathbf{v}} = \frac{\partial P}{\partial t} + \overline{\mathbf{v}} \cdot \nabla P + \nabla \cdot \overline{K} \nabla T, \tag{1.42}$$

using (1.17).

1.2.4 Temperature Formulation

We now connect enthalpy and temperature with thermodynamics relationships. To begin with, we write differential form of internal energy as

$$du_i = Tds_i - p_i d\tilde{V}_i. \tag{1.43}$$

We use (1.43) in the differential form of enthalpy as

$$dh_i = du_i + p_i d\tilde{V}_i + \tilde{V}_i dp_i = Tds_i - p_i d\tilde{V}_i + p_i d\tilde{V}_i + \tilde{V}_i dp_i = Tds_i + \frac{dp_i}{\rho_i}, \tag{1.44}$$

and entropy as

$$ds_i = c_p \frac{dT}{T} + \left(\frac{\partial s_i}{\partial p_i}\right)_T dp_i, \tag{1.45}$$

where c_p is the uniform specific heat capacity used for both phases. We replace $\left(\frac{\partial s}{\partial p}\right)_T$ with $-\left(\frac{\partial \tilde{V}_i}{\partial T}\right)_p$ according to Maxwell's fourth relation Campbell (1958). We assume a simple linear relationship between temperature and volume as

$$\frac{\partial \tilde{V}_i}{\partial T} = \alpha_\rho / \rho_i, \tag{1.46}$$

where α_ρ is the thermal expansion coefficient used uniformly for both the liquid and the solid phases. We substitute the entropy equation (1.45) into the enthalpy equation (1.44), so

$$dh_i = c_p dT + \frac{1 - \alpha_\rho T}{\rho_i} dp_i. \tag{1.47}$$

We can expand the divergence term in the equation (1.42) and write for phase i, $i \in \{s, l\}$ as

$$\nabla \cdot (\phi_i \rho_i h_i \mathbf{v}_i) = h_i \nabla \cdot (\phi_i \rho_i \mathbf{v}_i) + \phi_i \rho_i \mathbf{v}_i \nabla h_i, \tag{1.48}$$

where $\nabla h_i = c_p \nabla T + \frac{1-\alpha_\rho T}{\rho_i} \nabla p_i$. We now replace the term ∇h_i with the relation (1.46) as

$$\nabla \cdot \overline{\rho h \mathbf{v}} = (h_l - h_s) \nabla \cdot (\phi_l \rho_l \mathbf{v}_l) + h_s \nabla \cdot \overline{\rho \mathbf{v}} + c_p \overline{\rho \mathbf{v}} \cdot \nabla T + (1 - \alpha_\rho T) \overline{\mathbf{v}} \cdot \nabla p_i, \tag{1.49}$$

We substitute the new formulation back to the equation (1.42) and again use lithostatic pressure for approximation as

$$\frac{\partial \overline{\rho h}}{\partial t} + h_s \nabla \cdot \overline{\rho \mathbf{v}} + c_p \overline{\rho \mathbf{v}} \cdot \nabla T = -(h_l - h_s) \nabla \cdot (\phi_l \rho_l \mathbf{v}_l) + \frac{\partial P}{\partial t} + \nabla \cdot \overline{K} \nabla T + \alpha_\rho T \nabla P, \quad (1.50)$$

where we have $\nabla P = \rho \mathbf{g}$ under the lithostatic pressure approximation.

We integrate enthalpy (1.47) under constant pressure to relate enthalpy and temperature in the sense of thermodynamic state variables as

$$h - h_0 = c_p (T - T_0), \quad (1.51)$$

where the reference values h_0 and T_0 are set as zero for simplicity. Another important physical attribute in the context of melting is latent heat L. We write specific enthalpy in the solid and the liquid phase separately as

$$\begin{aligned} h_s &= c_p T, \\ h_l &= c_p T + L. \end{aligned} \quad (1.52)$$

Under the extended Boussinesq assumption, we rewrite the enthalpy equation with latent heat and temperature under constant pressure ($\partial P / \partial t = 0$) as

$$\frac{\partial \overline{\rho h}}{\partial t} + \rho c_p \nabla \cdot (\overline{\mathbf{v}} T) = \rho L \nabla \cdot (\phi_s \mathbf{v}_s) + \overline{K} \nabla^2 T + \rho \alpha_\rho T \overline{\mathbf{v}} \cdot \mathbf{g}, \quad (1.53)$$

since $\nabla \cdot (\phi_l \mathbf{v}_l) = -\nabla \cdot (\phi_s \mathbf{v}_s)$ using the continuous equation (1.17), i.e., $\nabla \cdot \overline{\mathbf{v}} = 0$. The last term including gravitational acceleration has previously been referred to as adiabatic cooling or decompression cooling by Mckenzie and Bickle (1988). Assuming no density difference between the solid and the liquid phase, we divide ρc_p on both sides of the equation to obtain a normalized form as

$$\frac{\partial \overline{H}}{\partial t} + \nabla \cdot (\overline{\mathbf{v}} T) = \frac{L}{c_p} \nabla \cdot (\phi_s \mathbf{v}_s) + \kappa \nabla^2 T + \frac{\alpha_\rho}{c_p} T \overline{\mathbf{v}} \cdot \mathbf{g}, \quad (1.54)$$

where $\overline{H} = \overline{\rho h} / \rho c_p = \overline{h} / c_p$ is the bulk specific enthalpy and $\kappa = \overline{K} / \rho c_p$ is the bulk heat diffusivity.

1.3 Eutectic Phase Behavior

With the transport equations defined, we can now proceed to redistribute chemical species and energy within the system. The next step is to determine the thermodynamic state of the system, assuming thermodynamic equilibrium is obtained at each time step. Recognizing that mid-ocean ridges form in regions where temperatures are generally lower, eutectic melting, where the presence of a secondary species significantly lowers the melting temperature, provides a suitable model for describing the melting behavior in these areas.

The binary eutectic system has been widely applied in studies involving chemical equilibrium, with detailed description can be found in textbooks such as Lupis (1983). In the context of mantle dynamics, eutectic melting was being incorporated by Ribe (1985). Moreover, eutectic phase behavior has been applied to solidification of alloys by Wheeler et al. (1996). More recently, binary eutectic melting has also been applied to water-salt-brine systems in glaciers Hesse et al. (2022).

We assume the two primary components in our partially molten system are olivine and orthopyroxene (opx), which are highly miscible in the liquid phase, but immiscible in the solid phase. Olivine is considered to dominate the system, while orthopyroxene is the minor component. Assuming thermodynamic equilibrium is obtained at each time step, we can determine the phase fraction of liquid, the volume fraction of each solid components and the temperature using a binary eutectic phase diagram with transported enthalpy and bulk composition.

A typical eutectic phase diagram is shown in Figure 1.2. There are two key concepts about the diagram:

- the eutectic melting point or eutectic temperature T_e, is the lowest temperature at which melting may occur. Melting at the eutectic temperature is called eutectic melting. The temperature of the system is fixed at the eutectic temperature when eutectic melting happens.

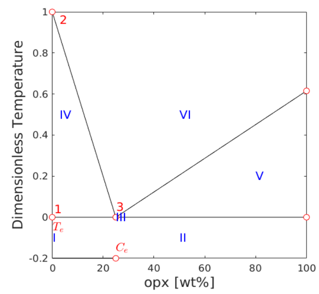


Figure 1.2: Eutectic phase diagram of a Olivine-Orthopyroxene system.

- The eutectic composition C_e is a fixed opx-olivine concentration ratio of the liquid produced by eutectic melting.

1.3.1 Clausius-Clapeyron Relation

In geological contexts, pressure can undergo significant variations during dynamic events such as mantle upwelling. As a result, the influence of pressure on the melting point cannot be neglected. To account for this, we approximate the solidus melting temperature T_m at a given pressure using the approximated Clausius–Clapeyron relationship. We also extend such approximation to eutectic temperature T_e as

$$\begin{aligned} T_m &= T_{m0} + \nu^{-1}(P - P_0), \\ T_e &= T_{e0} + \nu^{-1}(P - P_0), \end{aligned} \tag{1.55}$$

where ν is the Clausius-Clapeyron constant, and P_0 is the reference pressure usually set to be 0. The melting point T_{m0} and eutectic point T_{e0} refer to their respective values at standard atmospheric pressure.

1.3.2 Some Assumptions

In a eutectic system, the current thermodynamic state of the material can be identified using the bulk mass fractions of the components and bulk enthalpy of the system. Our bulk mass fraction will refer to opx in out binary system involving only two components so C is generally small. We begin by introducing some assumptions regarding densities and other material properties.

- We assume no density difference between solid phase olivine and opx, i.e., $\rho_{\text{opx}} = \rho_{\text{olivine}}$.
- We assume no density change during phase transition (extended Boussinesq approximation), i.e., $\rho_l = \rho_s$.
- We assume constant and uniform material thermodynamics properties like specific heat capacity c_p and thermal expansion coefficient α_p.

Following the assumptions, we relate the bulk composition with previously defined phase averaged concentration (1.39) as

$$\frac{\phi_l \rho_l c_l + \phi_s \rho_s c_s}{\phi_l \rho_l + \phi_s \rho_s} = \phi_l c_l + \phi_s c_s = \phi_2 + \phi_l c_l = C, \tag{1.56}$$

where ϕ_2 represents the phase fraction of solid opx. In the later discussion, ϕ_1 will be used to denote the phase fraction of solid olivine. We also express bulk specific enthalpy with temperature and latent heat as

$$\overline{H} = \phi_l h_l / c_p + \phi_s h_s / c_p = \phi_l T + \phi_s T + \phi_l L / c_p = T + \phi_l L / c_p. \tag{1.57}$$

According to equations (1.55), the difference between melting point and eutectic point, which is defined as $\Delta T = T_m - T_e = T_{m0} - T_{e0}$, remains constant regardless of pressure change. At any given pressure, we proceed by dividing by this ΔT to get a dimensionless formulation as

$$\frac{\overline{H}}{\Delta T} = \frac{T}{\Delta T} + \frac{\phi_l L}{c_p \Delta T}. \tag{1.58}$$

Let $\check{L} \coloneqq \frac{L}{c_p \Delta T}$, $H \coloneqq \frac{\overline{H}}{\Delta T}$ and $\check{T} \coloneqq \frac{T}{\Delta T}$. We write the dimensionless enthalpy temperature relationship as

$$H = \check{T} + \phi_l \check{L}. \tag{1.59}$$

1.3.3 Phase Diagram

Binary eutectic melting is usually highly-nonlinear. For simplicity, we adopt a geometric parameterization of the binary eutectic diagram that can be analytically evaluated. We implement a linear eutectic phase boundary to distinguish between different phase regions within the diagram.

The eutectic composition of opx is set as $C_e \approx 0.25$. For illustration only, we plot bulk composition against a renormalized dimensionless temperature $\check{T} - T_e/\Delta T$, where $\check{T}_m - T_e/\Delta T = 1$ and $\check{T}_e - T_e/\Delta T = 0$.

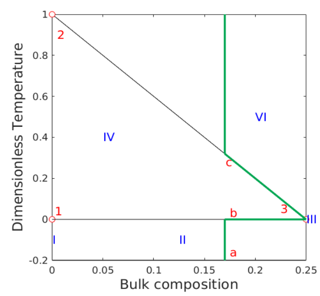


Figure 1.3

Figure 1.4: Phase diagram when the composition of opx is smaller than the eutectic composition. Region V does not appear.

We identify the following 6 regions with different phase assemblages:

I	Single-phase sub-solidus	: Olivine;
II	Two-phase sub-solidus	: Olivine + Orthopyroxene;
III	Three-phase eutectic	: Olivine + Orthopyroxene + melt;
IV	Two-phase super-eutectic	: Olivine + melt;
V	Two-phase super-eutectic	: Orthopyroxene + melt;
VI	Single-phase super-liquidus	: melt.

In a partial melting system, we assume the bulk composition of orthopyroxene (opx) remains below the eutectic composition. Consequently, we focus in the part of the phase diagram Figure 1.2, where the bulk composition is less than the eutectic composition, which is illustrated in the Figure 1.3.

Melting in the eutectic system is illustrated with the green line in the Figure 1.3. Now imagining a system containing 17% opx that starts to melting. Starting with all solid phase at Point a, the system temperature increase linearly in response to the enthalpy rise. When the temperature reaches the eutectic temperature, at Point b, both components of the system start melting with a fixed ratio of 25% opx and 75% olivine, which is the eutectic composition. Once all the solid phase opx is exhausted, the system leaves the eutectic points and solid olivine starts melting. The system climbs up along the line from Point 3 to Point c. After the remaining solid olivine has melted, the temperature of the system will continue to rise linearly in response to enthalpy input. It is worth noting that we can reverse the process to obtain a path of equilibrium crystallization.

1.3.4 Phase Split Calculation

A detailed description of the phase-splitting calculations is provided for the regions I, II, III, and IV in this chapter. Regions V and VI are not considered in this study, as we assume that the opx composition remains below the eutectic composition and that only partial melting occurs within the system. We draw a Enthalpy-Composition diagram including the effect of latent heat in the Figure 1.5. Again, we plot against the renormalized enthalpy $H - T_e/\Delta T$ for simplicity. The individual concentration and temperature are determined using equations (1.56) and (1.59). Follow the previous discussion, we denote the phase fraction of solid olivine as ϕ_1 and the phase fraction of solid opx as ϕ_2, such that $\phi_1 + \phi_2 = \phi_s$.

Region I. Pure olivine in the solid phase. The melting of pure olivine does not

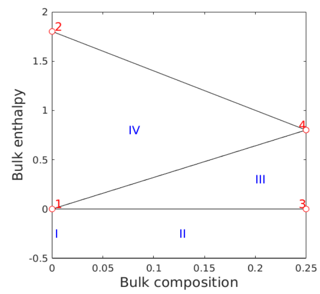


Figure 1.5: Enthalpy-Composition diagram

follow the eutectic melting path. The conditions to be in the region I are

$$C = 0, \quad H \leq \check{T}_m, \tag{1.60}$$

and we can determine the phase fractions and the temperature as

$$\phi_1 = 1, \quad \phi_2 = \phi_l = 0 \quad \text{and} \quad \check{T} = H. \tag{1.61}$$

We can determine partial derivatives of the temperature with respect to bulk enthalpy and bulk composition as

$$\frac{\partial \check{T}}{\partial H} = 1 \quad \text{and} \quad \frac{\partial \check{T}}{\partial C} = 0. \tag{1.62}$$

Region II. Sub-solidus: Olivine and opx coexist in the solid phase. In this phase region, temperature hasn't reached the eutectic melting temperature. The conditions to be in the region II are

$$C \neq 0, \quad H \leq \check{T}_e, \tag{1.63}$$

and we can determine the phase fractions and the temperature as

$$\phi_2 = C, \quad \phi_1 = 1 - \phi_2, \quad \phi_l = 0 \quad \text{and} \quad \check{T} = H. \tag{1.64}$$

We can determine partial derivatives of the temperature with respect to the bulk enthalpy and the bulk composition as

$$\frac{\partial \check{T}}{\partial H} = 1 \quad \text{and} \quad \frac{\partial \check{T}}{\partial C} = 0. \tag{1.65}$$

Region III. Two-phase eutectic: The system starts melting at eutectic temperature with a fixed liquid concentration. Both solid phase olivine and orthopyroxene and melt coexist at the same time. Temperature remains at the eutectic point until the opx is exhausted. Concentration of the opx in the melt is also fixed equal to the eutectic composition. The conditions to be in the region III are

$$C \leq C_e, \quad \check{T}_e < H \leq \mathrm{line}_{1-4}, \tag{1.66}$$

where line_{1-4} is defined as $H(C) = \check{T}_e + \check{L}C/C_e$. We can determine the phase fractions and the current temperature as

$$\phi_l = \frac{H - \check{T}_e}{\check{L}}, \quad \phi_2 = C - \phi_l C_e, \quad \phi_1 = 1 - \phi_l - \phi_2 \quad \text{and} \quad \check{T} = \check{T}_e. \tag{1.67}$$

We can determine the partial derivatives of the temperature with respect to the bulk enthalpy and the bulk composition as

$$\frac{\partial \check{T}}{\partial H} = 0 \quad \text{and} \quad \frac{\partial \check{T}}{\partial C} = 0. \tag{1.68}$$

Region IV. Super-eutectic two phase region: All solid phase opx has been exhausted. The concentration of opx in liquid c_l is now approximated with a linear function. The conditions to be in the region IV are

$$C \leq C_e, \quad \mathrm{line}_{1-4} < H \leq \mathrm{line}_{2-4}, \tag{1.69}$$

where line_{2-4} is defined as $H(C) = H_m - c_l/C_e$, and we can determine phase fractions as

$$\phi_2 = 0, \quad \phi_l = \frac{C}{c_l(\check{T})} \quad \text{and} \quad \phi_1 = 1 - \phi_l. \tag{1.70}$$

The corresponding temperature can be determined by solving the following equation

$$F(\check{T}) = H - \check{T} - \frac{C}{c_l(\check{T})}\check{L} = 0. \tag{1.71}$$

Let $c_l = C_e(\check{T}_m - \check{T})$, we can obtain a quadratic equation about temperature as

$$\check{T}^2 - (H + \check{T}_m)\check{T} + \check{T}_m H - CL/C_e = 0, \tag{1.72}$$

which can be solved with standard quadratic root finding formula. We omit the root that exceeding melting temperature, so

$$
\begin{aligned}
\check{T} &= \frac{H + \check{T}_m - \sqrt{(H + \check{T}_m)^2 - 4(\check{T}_m H - C\check{L}/C_e)}}{2} \\
&= \frac{H + \check{T}_m - \sqrt{(H - \check{T}_m)^2 + 4C\check{L}/C_e}}{2}.
\end{aligned} \tag{1.73}
$$

We can determine partial derivatives of the temperature with respect to the bulk enthalpy and the bulk composition as

$$
\begin{aligned}
\frac{\partial \check{T}}{\partial H} &= \frac{1}{2}\left(1 - \frac{H - \check{T}_m}{\sqrt{(H - \check{T}_m)^2 + 4C\check{L}/C_e}}\right), \\
\frac{\partial \check{T}}{\partial C} &= -\frac{\check{L}/C_e}{\sqrt{(H - \check{T}_m)^2 + 4C\check{L}/C_e}}.
\end{aligned} \tag{1.74}
$$

1.4 Nondimensionalization

Compaction and melting are the two most important aspects of mantle dynamics. We nondimensionlize the scaled formulation with a characteristic length scale related to compaction length and a scaled characteristic Darcy speed of the melt, i.e.,

$$
l_0 = \left(\frac{k_0 \mu_s}{\mu_l}\right)^{1/2} \quad \text{and} \quad u_0 = \frac{k_0 \rho_r g}{\mu_l}. \tag{1.75}
$$

We define a corresponding characteristic time scale as $t_0 = l_0/u_0$, and a characteristic pressure scale as $p_0 = \rho g l_0$. A set of nondimensional variables are defined as

$$
\check{\mathbf{v}}_r = \frac{\mathbf{v}_r}{u_0}, \quad \check{\mathbf{v}}_s = \frac{\mathbf{v}_s}{u_0}, \quad \check{q} = \frac{q}{p_0}, \quad \check{q}_l = \frac{q_l}{p_0}. \tag{1.76}
$$

The resulting dimensionless mechanics and transport equations are

$$
\check{\mathbf{v}}_r + \phi^{1+\Theta}\nabla(\phi^{-1/2}\check{q}_l) = 0, \tag{1.77}
$$

$$
\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_r) + \frac{1}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}) = 0, \tag{1.78}
$$

$$
\nabla\check{q} - \nabla\cdot\check{\tau}(\check{\mathbf{v}}_s) = -(1-\phi)\mathbf{n}, \tag{1.79}
$$

$$
\nabla\cdot\check{\mathbf{v}}_s - \frac{\phi^{1/2}}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}) = 0, \tag{1.80}
$$

Quantity	Value	Units
ρ	3000	$kg \cdot m^{-3}$
ρ_r	600	$kg \cdot m^{-3}$
α_ρ	3×10^{-5}	K^{-1}
c_p	1200	$J \cdot kg^{-1} \cdot K^{-1}$
L	5×10^5	$J \cdot kg^{-1}$
μ_l	1	Pascal $\cdot\, s$
μ_s	10^{19}	Pascal $\cdot\, s$
k_0	10^{-8}	m^2
T_{e0}	1480	K
T_{m0}	2000	K
ν	6.5×10^6	Pascal $\cdot\, K^{-1}$

Table 1.1: Flow Mechanics and Thermodynamics Parameters

where dimensionless deviatoric stress of the solid phase is $\check{\tau} = 2(1-\phi)(\frac{1}{2}(\nabla\check{\mathbf{v}}_s + \nabla\check{\mathbf{v}}_s^T) - 1/3\nabla \cdot \check{\mathbf{v}}_s\mathbf{I})$, which maintains the same form as (1.20), and $\mathbf{n}$ is the unit direction vector pointing downwards. Now we rewrite the transport equations with dimensionless variables as

$$\begin{aligned} \frac{\partial C}{\partial \check{t}} + \nabla \cdot (\check{\mathbf{v}}_{\text{eff}} C - \frac{\phi \mathfrak{D}_l}{u_0 l_0} \nabla c_l) = 0, \\ \frac{\partial H}{\partial \check{t}} + \nabla \cdot (\check{\bar{\mathbf{v}}} \check{T}) - \check{L} \nabla \cdot ((1-\phi)\check{\mathbf{v}}_s) - \frac{\kappa}{u_0 l_0} \nabla^2 \check{T} - \frac{\alpha_\rho l_0}{c_p} \check{T} \check{\bar{\mathbf{v}}} \cdot \mathbf{g} = 0. \end{aligned} \tag{1.81}$$

Parameters used in the flow mechanics and thermodynamic computations are listed in Table 1.1. The characteristic values are calculated as

$$l_0 = 3.16 \times 10^5 \text{m}, \quad u_0 = 5.88 \times 10^{-5} \text{m/s}, \quad t_0 = 5.38 \times 10^6 \text{s}, \quad \text{and} \quad p_0 = 9.30 \times 10^9 \text{Pascal}, \tag{1.82}$$

which are

$$l_0 = 316 \text{km}, \quad u_0 = 1.86 \text{km/yr}, \quad t_0 = 0.1705 \text{yrs}. \tag{1.83}$$

Chapter 2: Numerical Methods

In this chapter, we introduce a locally mass conserving numerical framework. A conforming mixed finite element method (MFEM) is proposed to solve the static Darcy-Stokes coupled methanics system. The proposed conforming MFEM captures the normal fluxes across element edges, ensuring compatibility with local mass conservation. The transport system is solved using a finite volume method (FVM), which inherently ensures local mass conservation.

We begin by introducing the mesh partition used for both the MFEM and the FVM schemes. Based on the mesh partition, two stable element pairs are presented, one for Darcy flow and one for Stokes flow. We then present a noval nonlinear WENO weighting scheme for the FVM. Error estimates are provided for both the MFEM and the FVM schemes.

2.1 Quadrilateral Mesh

We consider a bounded computational domain $\Omega \subset \mathbb{R}^2$, where $\partial\Omega$ denotes the Lipschitz-continuous boundary of the domain. For a given $h > 0$, let $\mathcal{T}_h$ be a *triangulation* of the closure of the domain $\overline{\Omega}$ made of quadrilaterals E, i.e.,

$$\overline{\Omega} = \cup_{E \in \mathcal{T}_h} E, \quad h_E < h, \tag{2.1}$$

where two quadrilaterals are either disjoint or share one edge or one vertex. We assume that the quadrilateral mesh is uniformly shape regular. A formal definition of uniform shape regularity of a quadrilateral mesh is given in Girault and Raviart (2011).

Definition 2.1. For any quadrilateral $E \in \mathcal{T}_h$, form the triangle $T_i, \quad i \in \{0, 1, 2, 3\}$ by excluding vertex v_i from the quadrilateral element E, and use the remaining three vertices. Let

$$\begin{aligned} h_E &= \text{diameter of Element } E, \\ \rho_E &= 2 \min_{1\le i\le 4}\{\text{diameter of largest circle inscribed in } T_i\}. \end{aligned} \tag{2.2}$$

A mesh partition $\mathcal{T}_h$ is considered to be uniformly shape regular if there exists a shape regularity parameter σ_* independent of $\mathcal{T}_h$, such that for all quadrilaterals $E \in \mathcal{T}_h$,

$$\rho_E/h_E \ge \sigma_* > 0. \tag{2.3}$$

Note that no subtriangle T_i of E has zero measure, i.e., each quadrilateral E is assumed to be strictly convex, not degenerating into a triangle, line or point.

The quadrilateral $E \in \mathcal{T}_h$ is referred to differently depending on the context. In the finite element method, a quadrilateral E is called an *element*, while in the finite volume method, it is typically referred to as a *cell*. For simplicity, we use the term *element* throughout this section.

In Figure 2.1, we show a reference element $\hat{E} = [-1, 1]^2$, a local reference element $\tilde{E}$ and a general quadrilateral element E. The edges of element E are denoted by e_i, $i \in \{0, 1, 2, 3\}$, and the vertices of the element E are denoted as v_i, $i \in \{0, 1, 2, 3\}$, both numbered in the counterclockwise order. We begin counting from zero, first to align with the indexing convention of C++, and second to facilitate representation in terms of modulo arithmetic, which also starts from zero. The vertices can be understood by $v_i = e_i \cap e_{i+1}$, where $i \equiv i \pmod 4$. Here, $\boldsymbol{\nu}_i$ and $\boldsymbol{\tau}_i$ are used to represent the unit normal and the unit tangent vector respectively. The unit normal vector $\boldsymbol{\nu}_i$ points outwards from the element E. The corresponding unit tangent $\boldsymbol{\tau}_i$ is obtained by rotating $\boldsymbol{\nu}_i$ counterclockwise, following the right-hand rule.

We assume the general quadrilateral element E can be obtained by scaling from the local reference element $\tilde{E}$ such that $\tilde{E} = E/H$, where H is the maximal edge length of E, i.e., the length of edge e_1 in Figure 2.1. Here, $|e_1| = H$ and $|\tilde{e}_1| = 1$. There exists a non-affine bijective and smooth map $\mathbf{F}_{\hat{E}} : \hat{E} \to \tilde{E}$ mapping the vertices of $\hat{E}$ to the vertices of $\tilde{E}$. Note that the unit normal and the unit tangent vectors are invariant under the scaling mapping H.

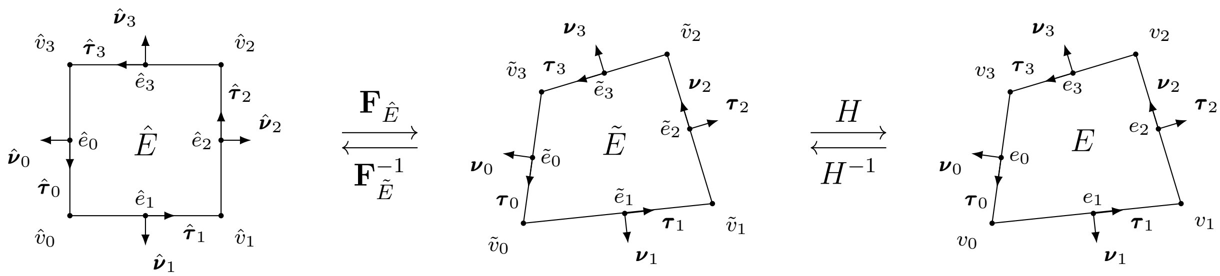


Figure 2.1: An exhibition of a reference square element, a local reference element, and a general non-degenerate quadrilateral element.

For the ease of implementation, we use a logically rectangular mesh, where each element is indexed using Cartesian coordinates $\{i, j\}$. In Figure 2.2, we illustrate a full logically rectangular, quasi-uniform quadrilateral mesh $\mathcal{T}_h$, which is generated by randomly distorting the interior vertices of a square mesh. The boundary vertices $v_i \in \partial\Omega$ remain fixed and are not subject to distortion.

2.2 Mathematical Preliminaries

In this section, we introduce some mathematical concepts that are relied upon in the remainder sections. The theory and notations are presented in a general dimension sense, but in practice, our discussion is confined to two dimension cases. In addition, for simplicity, we restrict our consideration to real-valued functions.

To begin with, we recall the definition of a Sobolev space.

Definition 2.2. Let $\Omega \subset \mathbb{R}^d$ be a domain, $1 \leq p \leq \infty$, and $m \geq 0$ be an integer. The Sobolev space of m derivatives in $L^p(\Omega)$ is

$$W_p^m(\Omega) = \{f \in L^p(\Omega) : D^\alpha f \in L^p(\Omega) \text{ for all multi-indices } \alpha \text{ such that } |\alpha| \leq m\}, \tag{2.4}$$

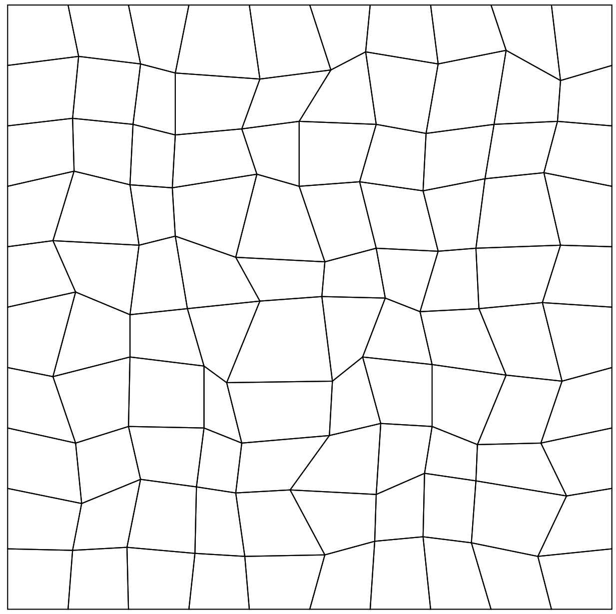


Figure 2.2: An exhibition of a full logically rectangular quasi-uniform quadrilateral mesh $\mathcal{T}_h$.

where the multi-index $\alpha = (\alpha_0, \alpha_1, \ldots, \alpha_{N-1}) \in \mathbb{N}^N$ and

$$|\alpha| = \sum_{i=0}^{N-1} \alpha_i.$$

The multi-index derivative is defined as

$$D^\alpha = \frac{\partial^{|\alpha|}}{\partial x_0^{\alpha_0} \ldots \partial x_{N-1}^{\alpha_{N-1}}}.$$

For $f \in L^p(\Omega)$, the $L^p(\Omega)$ norm is

$$\|f\|_{L^p(\Omega)} = \begin{cases} \left(\int_\Omega |f|^p \right)^{1/p}, & p < \infty, \\ \operatorname*{ess\,sup}_{x \in \Omega} |f(x)|, & p = \infty. \end{cases} \tag{2.5}$$

We proceed with the definition of the Sobolev norm and seminorm.

Definition 2.3. For $f \in W_p^m(\Omega)$, the norm $\|\cdot\|_{W_p^m}$ is

$$\|f\|_{W_p^m(\Omega)} = \left\{ \sum_{|\alpha| \leq m} \|D^\alpha f\|_{L^p(\Omega)}^p \right\}^{1/p}, \quad p < \infty, \tag{2.6}$$

and

$$\|f\|_{W^m_\infty(\Omega)} = \max_{|\alpha|\leq m} \|D^\alpha f\|_{L^\infty(\Omega)}, \quad p = \infty. \tag{2.7}$$

The seminorm $|\cdot|_{W^m_p}$ is

$$|f|_{W^m_p(\Omega)} = \left\{ \sum_{|\alpha|=m} \|D^\alpha f\|^p_{L^p(\Omega)} \right\}^{1/p}, \quad p < \infty. \tag{2.8}$$

In the special case when $p = 2$, we denote $W^m_2(\Omega) = H^m(\Omega)$. The corresponding norm and seminorn are then written as $\|\cdot\|_{H^m(\Omega)}$ and $|\cdot|_{H^m(\Omega)}$ respectively. Recall the Trace Theorem in $H^m(\mathbb{R}^d)$.

Theorem 2.1. *(Trace Theorem)*

Let k and d be integers with $0 < k < d$. The restriction map R extends to a bounded linear map from $H^m(\mathbb{R}^d)$ onto $H^{s-k/2}(\mathbb{R}^{d-k})$, provided that $s > k/2$.

Proof. See Arbogast and Bona (2025). □

Let $\mathbb{P}_r(\omega)$ denote the space of polynomials of degree up to r on $\omega \subset \mathbb{R}^d$. The dimension of $\mathbb{P}_r$ defined on $\mathbb{R}^d$ can be calculated as

$$\dim \mathbb{P}_r(\mathbb{R}^d) = \binom{r+d}{d} = \frac{(r+d)!}{r!d!}. \tag{2.9}$$

For simplicity, we momentarily restrict $d = 2$. Let $\mathbb{Q}_{r,s}(\omega) := \mathbb{P}_r(\omega) \otimes \mathbb{P}_s(\omega)$ denote the space of tensor product polynomials derived by polynomials of degree r in x and s in y on $\omega \subset \mathbb{R}^2$. The dimension of $\mathbb{Q}_{r,s}$ is

$$\dim \mathbb{Q}_{r,s}(\mathbb{R}^2) = \dim \mathbb{P}_r(\mathbb{R}) \times \dim \mathbb{P}_s(\mathbb{R}) = (r+1) \times (s+1). \tag{2.10}$$

We refer to the Bramble-Hilbert lemma for basic interpolation error estimation for polynomials of degree r.

Lemma 2.2. *(Bramble-Hilbert Lemma)*

Let $\Omega \in \mathbb{R}^d$ *be a domain, and function* $f \in W_p^{k+1}(\Omega)$, $k \geq 0$, $1 \leq p < \infty$. *There exists a constant* C *such that*

$$\inf_{p \in \mathbb{P}_k(E)} \|f - p\|_{W_p^m(E)} \leq C h^{k+1-m} |f|_{W_p^{k+1}(E)}, \quad m = 0, 1, \tag{2.11}$$

where the constant C *has an upper bound over all elements* $E \in \mathcal{T}_h$ *due to the shape regularity assumption.*

Proof. See Bramble and Hilbert (1970) and Dupont and Scott (1980). □

2.3 Direct Serendipity Space

Serendipity finite elements have the same accuracy of approximation with classical tensor product elements while using fewer degrees of freedom. However, the classical serendipity spaces suffer from poor approximation on quadrilaterals. To address this issue, we follow the approach of Arbogast et al. (2022) by defining local shape functions directly on quadrilateral elements, thereby recovering the desired accuracy. In the next sections, H^1 and $H(\text{div})$ conforming direct mixed finite element spaces are constructed with the direct serendipity function spaces.

To begin with, we introduce some auxiliary distance functions. We define the linear polynomial $\lambda_i(\mathbf{x})$ giving the distance of a point at coordinate $\mathbf{x} \in \mathbb{R}^2$ to edge e_i as

$$\lambda_i(\mathbf{x}) = -(\mathbf{x} - \mathbf{x}_i^*) \cdot \boldsymbol{\nu}_i, \quad i = 0, 1, 2, 3, \tag{2.12}$$

where $\mathbf{x}_i^* \in e_i$ is any point on edge e_i. We pick $\mathbf{x}_i^* = \mathbf{x}_{v,i}$ for simplicity, where $\mathbf{x}_{v,i}$ denotes the coordinates of corner v_i. The distance function λ_i returns a positive value as long as the point is located in the interior of the element E.

We define two additional distance functions giving the distance of a point at coordinate $\mathbf{x} \in \mathbb{R}^2$ to the diagonal line d_i joining v_i with v_{i+2}, $i \in 0, 1$. Let $\boldsymbol{\nu}_{d,i}$ be the unit normal vector to the diagonal line d_i. The direction of the $\nu_{d,i}$ is defined by rotating counterclockwise of the unit vector pointing from v_i to v_{i+2}. The resulting diagonal distance function is defined as

$$\lambda_{d,i}(\mathbf{x}) = -(\mathbf{x} - \mathbf{x}^*_{d,i}) \cdot \boldsymbol{\nu}_{d,i}, \quad i = 0, 1, \tag{2.13}$$

where $\mathbf{x}^*_{d,i}$ is any point on the diagonal line d_i. Here, we use the corner point v_i for simplicity, i.e., $\mathbf{x}^*_{d,i} = \mathbf{x}_{v,i}$.

We now present a set of nodal basis functions for the first and second order direct serendipity spaces, $\mathcal{DS}_1$ and $\mathcal{DS}_2$.

2.3.1 Second-order Space $\mathcal{DS}_2$

The dimension of the second-order polynomials in two dimension is calculated by (2.9) as

$$\dim \mathbb{P}_2(\mathbb{R}^2) = \binom{2+2}{2} = \frac{(4)!}{2!2!} = 6, \tag{2.14}$$

while a quadrilateral element requires a minimum of 4 vertex dofs and 4 edge dofs. Therefore, we supplement polynomial space $\mathbb{P}_2(E)$ with two linearly independent functions, $\phi_{s,1}(\mathbf{x})$ and $\phi_{s,2}(\mathbf{x})$, spanning the supplemental function space $\mathbb{S}^{\mathcal{DS}}_r(E) = \operatorname{span}\{\phi_{s,1}, \phi_{s,2}\}$, $r \geq 2$. We present the second-order direct serendipity space as

$$\mathcal{DS}_2(E) = \mathbb{P}_2(E) \oplus \mathbb{S}^{\mathcal{DS}}_2(E). \tag{2.15}$$

The supplemental functions are defined as

$$\phi_{s,i}(\mathbf{x}) = \lambda_{i+1}(\mathbf{x})\lambda_{i+2}(\mathbf{x})R_{i,i+2}(\mathbf{x}), \quad i \in \{0, 1\}. \tag{2.16}$$

We use a simple definition for rational function $R_{i,i+2}$ as

$$R_{i,i+2}(\mathbf{x}) = \frac{\lambda_i(\mathbf{x}) - \lambda_{i+2}(\mathbf{x})}{\lambda_i(\mathbf{x}) + \lambda_{i+2}(\mathbf{x})}, \tag{2.17}$$

which satisfies the conditions

$$R_{i,i+2}|_{e_i} = -1, \quad R_{i,i+2}|_{e_{i+2}} = 1. \tag{2.18}$$

We continue to define

$$R_i(\mathbf{x}) = \frac{1}{2}\Big(1 - R_{i,i+2}(\mathbf{x})\Big) = \frac{\lambda_{i+2}(\mathbf{x})}{\lambda_i(\mathbf{x}) + \lambda_{i+2}(\mathbf{x})}, \quad i \in \{0,1,2,3\}, \tag{2.19}$$

where $i \equiv i \pmod 4$ as usual, and $R_{i,i+2} = -R_{i+2,i}$ according to the equation (2.17). Here, $R_i|_{e_i} = 1$, and $R_i|_{e_{i+2}} = 0$ by definition. In Figure 2.3, we draw R_0 on a general quadrilateral element, which evaluates to 1 on edge e_0, and evaluates to 0 on the opposite edge e_2.

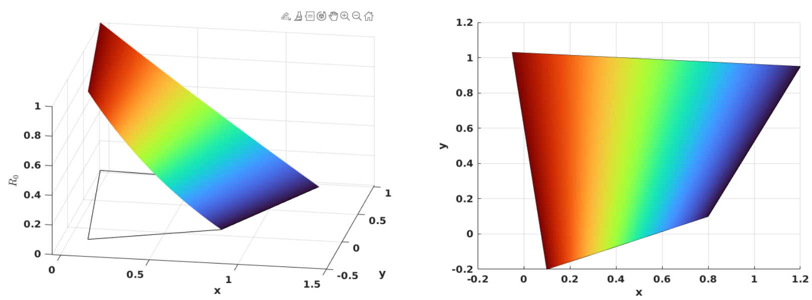


Figure 2.3: Rational function R_0.

The edge nodal basis functions for $\mathcal{DS}_2$ are constructed as

$$\varphi^{(2)}_{e,i}(\mathbf{x}) = \frac{\lambda_{i+1}(\mathbf{x})\lambda_{i+3}(\mathbf{x})R_i(\mathbf{x})}{\lambda_{i+1}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})R_i(\mathbf{x}_{e,i})}, \tag{2.20}$$

where $\mathbf{x}_{e,i}$ are the coordinates of the middle point of the edge e_i, again $i \equiv i \pmod 4$. We use superscript (2) to denote the order of the function space. We can restrict the edge basis function to edges as

$$\varphi^{(2)}_{e,i}|_{e,j} = \begin{cases} \dfrac{\lambda_{i+1}(\mathbf{x})\lambda_{i+3}(\mathbf{x})}{\lambda_{i+1}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})}, & i = j, \\ 0, & i \neq j. \end{cases} \tag{2.21}$$

In Figure 2.4, we present edge basis function $\varphi^{(2)}_{e,0}$ on a general quadrilateral element, where $\varphi^{(2)}_{e_0}$ evaluates 0 on edges $\{e_i, \quad i \neq 0\}$, and evaluates to a quadratic function on edge e_0.

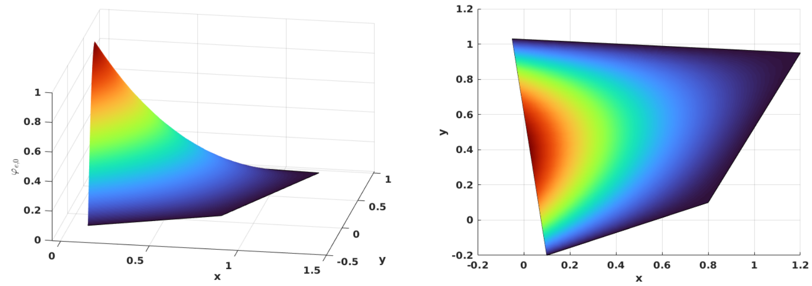


Figure 2.4: Edge nodal basis function $\varphi_{e,0}^{(2)}$ in $\mathcal{DS}_2$.

The vertex nodal basis functions for $\mathcal{DS}_2$ are constructed as

$$\varphi_{v,i}^{(2)}(\mathbf{x}) = \frac{\lambda_{i+2}(\mathbf{x})\lambda_{i+3}(\mathbf{x}) - \sum_{k=i}^{i+1} \lambda_{i+2}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})\varphi_{e,i}^{2}(\mathbf{x})}{\lambda_{i+2}(\mathbf{x}_{v,i})\lambda_{i+3}(\mathbf{x}_{v,i}) - \sum_{k=i}^{i+1} \lambda_{i+2}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})\varphi_{e,i}^{(2)}(\mathbf{x}_{v,i})}, \tag{2.22}$$

where $\mathbf{x}_{v,i}$ is the coordinates of the vertex node v_i, and $i \equiv i \pmod 4$ as well. At the nodal points

$$\varphi_{v,i}^{(2)}(\mathbf{x}) = \begin{cases} 1, & \mathbf{x} = \mathbf{x}_{v,i}, \\ 0, & \mathbf{x} = \mathbf{x}_{v,j} \text{ or } \mathbf{x}_{e,k}, \quad j \neq i, \ \forall k. \end{cases} \tag{2.23}$$

In Figure 2.5, we show $\varphi_{v,0}^{(2)}$ on a general quadrilateral.

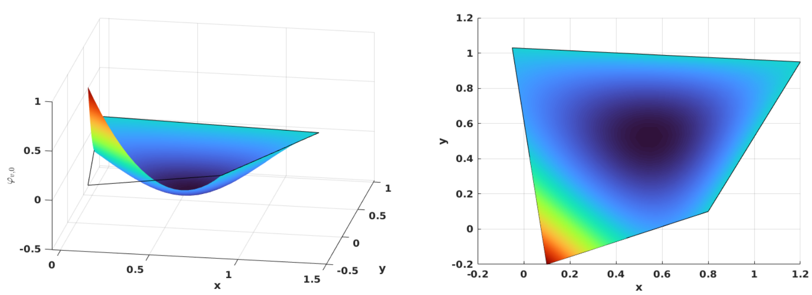


Figure 2.5: Vertex nodal basis function $\varphi_{v,0}^{2}$ in $\mathcal{DS}_2$.

We conclude that unisolvence follows directly from the constructed nodal basis functions.

2.3.2 First-order Space $\mathcal{DS}_1$

The dimension of the first-order polynomials in two dimension is also calculated by equation (2.9) as

$$\dim \mathbb{P}_1(\mathbb{R}^2) = \binom{1+2}{2} = \frac{(3)!}{2!1!} = 3, \tag{2.24}$$

while a quadrilateral element requires a minimum of 4 vertex dofs. Therefore, we supplement polynomial space $\mathbb{P}_1(E)$ with one additional function. The supplemental function for $\mathcal{DS}_1$ differs from that for $\mathcal{DS}_2$. We present the first order direct serendipity space as

$$\mathcal{DS}_1(E) = \mathbb{P}_1(E) \oplus \operatorname{span}\{R\}, \tag{2.25}$$

where $R(\mathbf{x}_{v,0}) = R(\mathbf{x}_{v,2}) = -R(\mathbf{x}_{v,1}) = -R(\mathbf{x}_{v,3}) = 1$. The construction of the rational function R is not unique. We present one way to construct $R(\mathbf{x})$ using edge nodal basis function (2.20) in $\mathcal{DS}_2$ as

$$R(\mathbf{x}) = r(\mathbf{x}) - \sum_{i=0}^{3} r(\mathbf{x}_{e,i})\varphi_{e,i}^{(2)}(\mathbf{x}), \tag{2.26}$$

where the auxiliary function $r(\mathbf{x})$ is defined as

$$r(\mathbf{x}) = \frac{\lambda_2(\mathbf{x})\lambda_3(\mathbf{x})}{\lambda_2(\mathbf{x}_{v,0})\lambda_3(\mathbf{x}_{v,0})} - \frac{\lambda_0(\mathbf{x})\lambda_3(\mathbf{x})}{\lambda_0(\mathbf{x}_{v,1})\lambda_3(\mathbf{x}_{v,1})} + \frac{\lambda_0(\mathbf{x})\lambda_1(\mathbf{x})}{\lambda_0(\mathbf{x}_{v,2})\lambda_1(\mathbf{x}_{v,2})} - \frac{\lambda_1(\mathbf{x})\lambda_2(\mathbf{x})}{\lambda_1(\mathbf{x}_{v,3})\lambda_2(\mathbf{x}_{v,3})}. \tag{2.27}$$

In Figure 2.6, we plot the rational function $R(\mathbf{x})$ on a general quadrilateral.

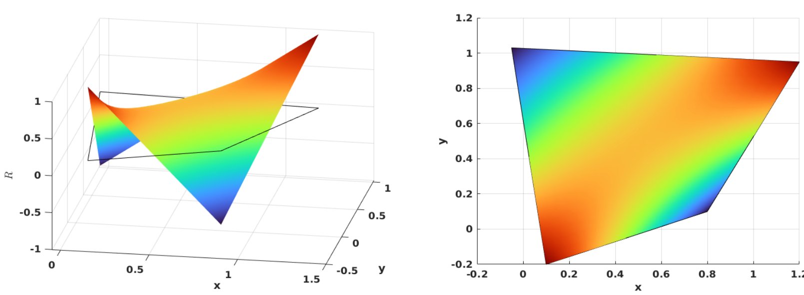


Figure 2.6: Supplemental rational function R in $\mathcal{DS}_1$.

We then define vertex nodal basis function accordingly as

$$\varphi_{v,i}^{(1)}(\mathbf{x}) = \frac{\lambda_{d,m}(\mathbf{x}) - \frac{1}{2}\lambda_{d,m}(\mathbf{x}_{v,i+2})(1+(-1)^i R(\mathbf{x}))}{\lambda_{d,m}(\mathbf{x}_{v,i}) - \lambda_{d,m}(\mathbf{x}_{v,i+2})}, \tag{2.28}$$

where $m \equiv i+1 \pmod 2$, and $i \equiv i \pmod 4$. In Figure 2.7, we show the vertex nodal basis function $\varphi_{v,0}^{(1)}$ in $\mathcal{DS}_1$ on a general quadrilateral, which evaluates 1 at vertex v_0 and 0 at other vertices.

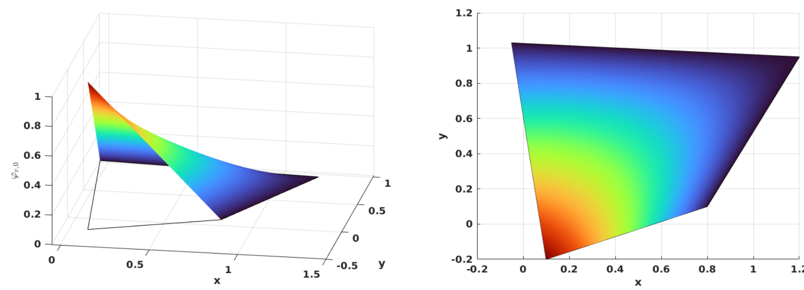


Figure 2.7: Vertex nodal basis function in $\mathcal{DS}_1$.

2.3.3 An Interpolation Operator

For completeness, we describe an interpolation operator mapping onto $\mathcal{DS}_r(K_i)$, $r = 1, 2$, following Arbogast et al. (2022). For each node a_i, the associated K_i has different meanings depending on the type of the nodal basis functions. If the basis function is associated with an edge node, we set $K_i = e_i$, where e_i is the corresponding edge. For the basis function associated with a vertex node, we set K_i to be any edge e containing node a_i, with the restriction that if $a_i \in \partial\Omega$, then $e \in \partial\Omega$.

An interpolation operator $\mathcal{I}_h^r : W_p^l(\Omega) \to \mathcal{DS}_r$ is defined as

$$\mathcal{I}_h^r v(\mathbf{x}) = \sum_{i=1}^{N_r} \varphi_i(\mathbf{x}) \int_{K_i} \psi_i(\mathbf{y}) v(\mathbf{y})\, d\mathbf{y} \in \mathcal{DS}_r, \tag{2.29}$$

where $1 \le p \le \infty$ and $l > 1/p$ (but $l \ge 1$ if $p = 1$), and $\psi(\mathbf{x})$ is the dual nodal basis such that

$$\int_{K_i} \psi_i(\mathbf{x}) \varphi_j(\mathbf{x})\, d\mathbf{x} = \delta_{i,j}, \quad i, j \in \{0, 1, 2, \ldots, N_r\}, \tag{2.30}$$

where N_r is the number of nodal basis functions. We present the following two lemmas regarding the constructed interpolation operator.

Lemma 2.3. *(Boundedness of the interpolation operator.)*

Let $\mathfrak{T}_h$ be uniformly shape regular according to Definition 2.3 with shape regularity parameter σ_*. Let $v \in W_p^l(\Omega)$, with $1 \leq p \leq \infty$ and $l > q/p$. Moreover, assume that $R_{i,i+2}$ is a uniformly differentiable functions of the vertices of E up to order m. Then for $r = 1, 2$, $E \in \mathfrak{T}_h$, $1 \leq q \leq \infty$, and any non negative integer m,

$$\|\mathfrak{I}_h^r v\|_{W_q^m(E)} \leq C(\sigma_*, m, q) \sum_{k=0}^{l} h_E^{k-m+\frac{2}{q}-\frac{2}{p}} |v|_{W_p^k(E^*)}, \tag{2.31}$$

where $E^* = \cup_{F \in \mathfrak{T}_h, F \cap E \neq \emptyset} F$ and $|\cdot|_{W_p^k}$ is the seminorm of k-th order derivatives.

Proof. See Arbogast et al. (2022). □

Combining Lemma 2.3 and the Bramble-Hilbert lemma 2.2, we obtain local and global error estimation.

Lemma 2.4. *(Approximability of the interpolation operator.)*

Under the assumption of Lemma 2.3, there exists a constant $C = C(r, \sigma_*) > 0$ such that for all functions $v \in W_{l,p}(E^*)$, with $1 \leq p \leq \infty$ and $l > 1/p$ or ($l \geq 1$ if $p = 1$),

$$\|v - \mathfrak{I}_h^r v\|_{W_p^m(E)} \leq C h_E^{l-m} |v|_{W_p^l(E^*)}, \quad 0 \leq m \leq \min(l, r+1). \tag{2.32}$$

Moreover, there exists a constant $C = C(r, \sigma_*) > 0$, independent of $h = \max_{E \in \mathfrak{T}} h_E$, such that for all functions $v \in W_p^l(\Omega)$, $p < \infty$

$$\left(\sum_{E \in \mathfrak{T}_h} \|v - \mathfrak{I}_h^r v\|_{W_p^m(E)}^p \right)^{1/p} \leq C h^{l-m} |v|_{W_p^l(\Omega)}, \quad 0 \leq m \leq \min(l, r+1). \tag{2.33}$$

Proof. See Arbogast et al. (2022). □

2.4 Mixed Finite Elements

In this section, we describe our mixed finite element approximation for the scaled coupled system (1.77)–(1.80). We begin by presenting the mixed variational formulations of the Stokes and Darcy model problems separately. Subsequently, we introduce H^1- and $H(\text{div})$-conforming mixed finite element spaces. We describe the construction of mixed finite elements from the aforementioned direct serendipity space. Recall Ciarlet's definition Ciarlet (2002) of a finite element.

Definition 2.4. A triple $(E, X(E), \{\phi_1, \ldots, \phi_n\})$ defines a *finite element*, where

- The element $E \in \mathcal{T}_h$;
- The space of shape functions is $X(E)$ with basis $\{\varphi_1, \ldots, \varphi_n\}$,where the dimension of the shape function space is $\dim(X(E)) = n$;
- A set of continuous and linear functionals is $\{\phi_1, \ldots, \phi_n\}$, defined on a subset $\mathcal{X}(E)$ of the energy space containing the finite element space $X(E)$ as

$$\phi_j : \mathcal{X}(E) \to \mathbb{R}, \tag{2.34}$$

such that the unisolvence condition is achieved as

$$\langle \phi_j, \varphi_i \rangle = \delta_{ij}, \quad i, j \in \{1, 2, 3, \ldots, n\}. \tag{2.35}$$

In later subsections, we give an explicit construction of the nodal basis functions and degrees of freedom (values of the $\phi_j,\ \ j = 1, 2, \ldots, n$) that is associated to them, thereby completing the definition of the finite element.

2.4.1 A Darcy Model Problem

Consider a model problem for Darcy flow equipped with homogeneous boundary condition as

$$\begin{aligned} \mathbf{u} + \nabla p &= \mathbf{f}, \ \text{ in } \Omega, \\ \nabla \cdot \mathbf{u} &= 0, \ \text{ in } \Omega, \\ \mathbf{u} \cdot \boldsymbol{\nu} &= 0, \ \text{ on } \partial\Omega. \end{aligned} \tag{2.36}$$

We seek the displacement and pressure solution pair $(\mathbf{u}, p)$ in the identical test and trial spaces

$$\begin{aligned}
\mathbb{V}_D &= \{\mathbf{v} \in (H(\text{div};\Omega))^2 : \mathbf{v}\cdot\boldsymbol{\nu} = \mathbf{0} \ \text{ on } \partial\Omega\},\\
\mathbb{U}_D &= \{\mathbf{u} \in (H(\text{div};\Omega))^2 : \mathbf{u}\cdot\boldsymbol{\nu} = \mathbf{0} \ \text{ on } \partial\Omega\} = \mathbb{V}_D,\\
\mathbb{W}_D &= \left\{p \in L^2(\Omega) : \int_\Omega p\,d\mathbf{x} = 0\right\} = L^2(\Omega)/\mathbb{R}.
\end{aligned} \tag{2.37}$$

The standard mixed weak form reads as follows:

Find $\mathbf{u} \in \mathbb{U}_D$ and $p \in \mathbb{W}_D$ such that

$$\begin{aligned}
(\mathbf{u}, \mathbf{v}) - (p, \nabla\cdot\mathbf{v}) &= (\mathbf{f}, \mathbf{v}), \quad \forall \mathbf{v} \in \mathbb{V}_D,\\
(\nabla\cdot\mathbf{u}, \omega) &= 0, \qquad\quad \forall \omega \in \mathbb{W}_D.
\end{aligned} \tag{2.38}$$

We introduce two finite-dimensional subspaces $\mathbb{V}_{D,h} \subset \mathbb{V}_D$ and $\mathbb{W}_{D,h} \subset \mathbb{W}_D$, and consider the discretized problem:

Find $(\mathbf{u}_h, p_h) \in \mathbb{V}_{D,h} \times \mathbb{W}_{D,h}$ such that

$$\begin{aligned}
(\mathbf{u}_h, \mathbf{v}_h) - (p_h, \nabla\cdot\mathbf{v}_h) &= (\mathbf{f}, \mathbf{v}_h), \quad \forall \mathbf{v}_h \in \mathbb{V}_{D,h},\\
(\nabla\cdot\mathbf{u}_h, \omega_h) &= 0, \qquad\quad \forall \omega_h \in \mathbb{W}_{D,h}.
\end{aligned} \tag{2.39}$$

In the next subsection, we present a construction of the discretized space pair $\mathbb{V}_{D,h} \times \mathbb{W}_{D,h}$ for this Darcy model problem.

2.4.2 A reduced $H(\mathbf{div})$ conforming space $\mathbb{V}_1^0$

We have some choice to discretize the space $\mathbb{V}_D$. For example, we can choose the classical Ravier-Thomas element Raviart and Thomas (1977). The serendipity space is the natural precursor of the Brezzi-Douglas-Marini element Brezzi et al. (1985) related by a de Rham complex. We generate a reduced first-order $H(\text{div})$ approximation from the direct serendipity space, where $\nabla\cdot\mathbf{u}$ is approximated one less order than approximation of $\mathbf{u}$. We can also recover the Arbogast-Correa element Arbogast and Correa (2016) if we use the mapped serendipity space, as defined in that paper.

We write the de Rham exact sequence for the related energy spaces as

$$\begin{array}{ccccccc} \mathbb{R} \xhookrightarrow{\text{i}} & H^1 & \xrightarrow{\text{curl}} & H(\text{div}) & \xrightarrow{\text{div}} & L^2 & \xrightarrow{0} 0 \\ & \cup & & \cup & & \cup & \\ & \mathcal{DS}_2 & \xrightarrow{\text{curl}} & \mathbf{V}_1^0 & \xrightarrow{\text{div}} & \mathbb{P}_0 & . \end{array} \tag{2.40}$$

We define the reduced first-order $H(\text{div})$ conforming space as the first-order polynomial space supplemented with the curl of the second-order direct serendipity supplemental space of $\mathcal{DS}_2$, i.e.,

$$\mathbb{V}_1^0(E) = (\mathbb{P}_1(E))^2 \oplus \text{curl}\, \mathbb{S}_2^{\mathcal{DS}}(E). \tag{2.41}$$

Here, $\mathbb{V}_1^0(E)$ can be given a global basis with the normal components of the shape functions merged continuously on the shared edge of two elements.

The dimension of this reduced first-order $H(\text{div})$ conforming space is

$$\dim(\mathbb{V}_1^0(E)) = 2 \times 3 + 2 = 8. \tag{2.42}$$

On each edge of a quadrilateral element E, two independent basis functions are defined: one with vanishing average normal flux across the edge, and the other with non-vanishing average flux.

We state the degrees of freedom as normal fluxes on each edge as

$$DOF_{e_i}(\lambda_1) = \langle \mathbf{u}_h \cdot \boldsymbol{\nu}_i, \lambda_1 \rangle_{e_i}, \quad \lambda_1 \in \mathbb{P}_1, \tag{2.43}$$

where λ_1 is a linear function restricted to edge e_i, and $\dim(\mathbb{P}_1) = 2$. We then present a construction of two independent shape functions on each edge: $\boldsymbol{\varphi}_{l,i}$ and $\boldsymbol{\varphi}_{c,i}$, such that

$$\boldsymbol{\varphi}_{c,i} \cdot \boldsymbol{\nu}_j|_{e_j} = \begin{cases} 0, & j \neq i, \\ 1, & j = i. \end{cases} \tag{2.44}$$

$$\boldsymbol{\varphi}_{l,i} \cdot \boldsymbol{\nu}_j|_{e_j} = \begin{cases} 0, & j \neq i, \\ l_i(\mathbf{x}), & j = i, \end{cases} \tag{2.45}$$

where l_i is a non vanishing linear polynomial defined on edge e_i orthogonal to one, i.e., $\int_{e_i} l_i \, ds = 0$.

To begin with, we take the curl of the auxiliary rational function R_i (2.17) as

$$\operatorname{curl} R_i(\mathbf{x}) = \frac{\boldsymbol{\tau}_{i+2}(\mathbf{x})\lambda_i(\mathbf{x}) - \boldsymbol{\tau}_i(\mathbf{x})\lambda_{i+2}(\mathbf{x})}{(\lambda_i(\mathbf{x}) + \lambda_{i+2}(\mathbf{x}))^2}, \tag{2.46}$$

where $i \equiv i \pmod 4$. We present one way to construct the "linear" shape function $\boldsymbol{\varphi}_{l,i}$ in two steps as

$$\tilde{\boldsymbol{\varphi}}_{l,i} = \operatorname{curl}(\lambda_{i+1}\lambda_{i+3}R_i) = (\boldsymbol{\tau}_{i+1}\lambda_{i+3} + \lambda_{i+1}\boldsymbol{\tau}_{i+3})R_i + \lambda_{i+1}\lambda_{i+3}\operatorname{curl}R_i. \tag{2.47}$$

We then normalize it such that l_i varies from -1 to 1 when restricted to edge e_i as

$$\boldsymbol{\varphi}_{l,i} = \frac{\tilde{\boldsymbol{\varphi}}_{l,i}(\mathbf{x})|e_i|}{4\lambda_{i+1}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})R_i(\mathbf{x}_{e,i})}, \tag{2.48}$$

where the index $i \equiv i \pmod 4$, and the length of edge e_i is $|e_i| = |\mathbf{x}_{v,i} - \mathbf{x}_{v,i+3}|$. In Figure 2.8, we draw $\boldsymbol{\varphi}_{l,2}$ on the left element and $\boldsymbol{\varphi}_{l,0}$ on the right element sharing an edge, and show the normal flux, but not the tangential flux joining continuously as a linear function on the shared edge.

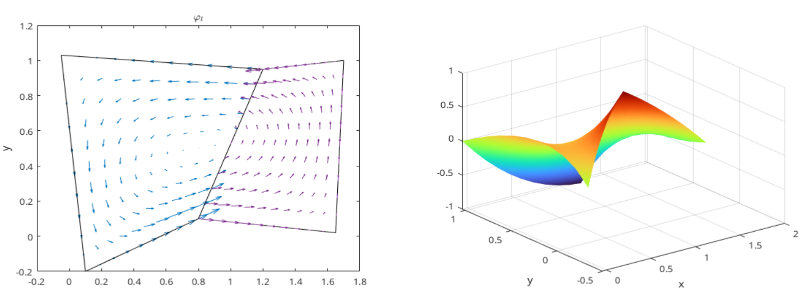


Figure 2.8: "Linear" shape function $\boldsymbol{\varphi}_l$.

The construction of the constant basis functions is a bit more complicated. We follow the process by Arbogast et al. (2022). We start by defining a linear nodal function $\phi^*_{v,i} \in \mathcal{DS}_2(E)$ such that $\phi^*_{v,i}(\mathbf{x}_{v,k}) = \delta_{ik}$. Its restriction to each edge of E is a linear function as

$$\phi^*_{v,i}(\mathbf{x}) = \varphi^2_{v,i}(\mathbf{x}) + \frac{1}{2}\big[\varphi^2_{e,i} + \varphi^2_{e,i+1}\big], \tag{2.49}$$

where $i \equiv i \pmod 4$. We then define $\boldsymbol{\varphi}^*_{v,i}(\mathbf{x}) = \operatorname{curl} \phi^*_{v,i}$ so that

$$\boldsymbol{\varphi}^*_{v,i} \cdot \boldsymbol{\nu}_j|_{e,j} = \begin{cases} 1/|e_i|, & j = i, \\ -1/|e_i|, & j = i+1, \\ 0, & j = i+2, i+3. \end{cases} \tag{2.50}$$

The "constant" basis function on the edge can now be defined as

$$\boldsymbol{\varphi}_{c,i} = \frac{(\mathbf{x}_{v,i} - \mathbf{x}_{v,i+2}) \cdot \boldsymbol{\nu}_{i+1}|e_{i+1}|\boldsymbol{\varphi}^*_{v,i} + \mathbf{x} - \mathbf{x}_{v,i+2}}{(\mathbf{x}_{v,i} - \mathbf{x}_{v,i+2}) \cdot \boldsymbol{\nu}_{i+1}|e_{i+1}|/|e_i| + (\mathbf{x}_{v,i} - \mathbf{x}_{v,i+2}) \cdot \boldsymbol{\nu}_i}, \tag{2.51}$$

where $i \equiv i \pmod 4$. The "constant" shape function $\boldsymbol{\varphi}_{c,i}$ has a normal flux of constant value 1 on edge e_i and evaluates to 0 normal flux on all the other edges. In Figure 2.9, we draw $\boldsymbol{\varphi}_{c,2}$ on the left element and $\boldsymbol{\varphi}_{c,0}$ on the right element sharing an edge, and show that the normal flux joins continuously as a constant 1 on the shared edge.

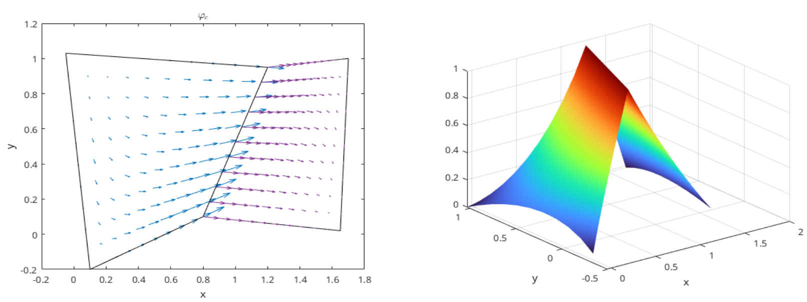


Figure 2.9: "Constant" shape function $\boldsymbol{\varphi}_c$.

2.4.3 Stability and Approximation Properties for $\mathbb{V}^0_1$

The accuracy and stability of $\mathbb{V}^0_1$ has been well studied and tested in the work by Arbogast et al. (2022). We state the result directly for completeness. They define a projection operation π that preserves element average divergence and average edge normal fluxes,

$$\pi : H(\mathrm{div};\Omega) \cap (L^{2+\epsilon}(\Omega))^2 \to \mathbb{V}^0_1, \tag{2.52}$$

where $\epsilon > 0$. This global projection operator π is a combination of local operators π_E defined on each element E according to the degrees of freedom. The operator π

also satisfies the commuting diagram property

$$\mathcal{P}_{W_0} \nabla \cdot \mathbf{v} = \nabla \cdot \pi \mathbf{v}, \tag{2.53}$$

where $\mathcal{P}_{W_0}$ is the L^2-orthogonal projection operator onto $W_0 = \nabla \cdot \mathbb{V}_1^0$, which is a space of piecewise constant values.

Lemma 2.5. *Stability and approximation properties for $\mathbb{V}_1^0$.*

Under the assumptions of Lemma 2.3 and Lemma 2.2, there exists a constant $C > 0$, independent of $\mathcal{T}_h$ and $h > 0$, such that

$$\begin{aligned} \|\mathbf{u} - \pi_D \mathbf{u}\|_{0,\Omega} &\leq C \|\mathbf{u}\|_{k,\Omega} h^k, && k = 1, 2, \\ \|\nabla \cdot (\mathbf{u} - \pi_D \mathbf{u})\|_{0,\Omega} &\leq C \|\nabla \cdot \mathbf{u}\|_{k,\Omega} h^k, && k = 0, 1, \\ \|p - \mathcal{P}_{W_s} p\|_{0,\Omega} &\leq C \|p\|_{k,\Omega} h^k, && k = 0, 1. \end{aligned} \tag{2.54}$$

Moreover, the discrete inf-sup condition holds

$$\sup_{\mathbf{v}_h \in \mathbb{V}_1^0} \frac{(\omega_h, \nabla \cdot \mathbf{v}_h)}{\|\mathbf{v}_h\|_{H(div)}} \geq \gamma \|\omega_h\|_{0,\Omega}, \quad \forall \omega_h \in W_0, \tag{2.55}$$

for some $\gamma > 0$ independent of $\mathcal{T}_h$ and $h > 0$.

Proof. See Arbogast et al. (2022). □

2.4.4 A Stokes Model Problem

Consider a model problem of an incompressible isotropic equation equipped with homogeneous boundary condition as

$$\begin{aligned} -\Delta \mathbf{u} + \nabla p &= \mathbf{f} \quad \text{in } \Omega, \\ \nabla \cdot \mathbf{u} &= 0 \quad \text{in } \Omega, \\ \mathbf{u} &= \mathbf{0} \quad \text{on } \partial\Omega. \end{aligned} \tag{2.56}$$

We choose energy spaces for the displacement-pressure pair $(\mathbf{u}, p)$ in the identical test and trial spaces as

$$\begin{aligned} \mathbb{V}_S &= \{\mathbf{v} \in (H^1(\Omega))^2 : \mathbf{v} = \mathbf{0} \text{ on } \partial\Omega\} = (H_0^1(\Omega))^2, \\ \mathbb{U}_S &= \{\mathbf{u} \in (H^1(\Omega))^2 : \mathbf{u} = \mathbf{0} \text{ on } \partial\Omega\} = \mathbb{V}_S, \\ \mathbb{W}_S &= \left\{ p \in L^2(\Omega) : \int_\Omega p \, d\mathbf{x} = 0 \right\} = L^2(\Omega)/\mathbb{R}. \end{aligned} \tag{2.57}$$

We consider the standard mixed variational form as

Find $\mathbf{u} \in \mathbb{U}_S$ and $p \in \mathbb{W}_S$ such that

$$\begin{aligned} (\nabla \mathbf{u}, \nabla \mathbf{v}) - (p, \nabla \cdot \mathbf{v}) &= (\mathbf{f}, \mathbf{v}), \quad \forall \mathbf{v} \in \mathbb{V}_S, \\ (\nabla \cdot \mathbf{u}, \omega) &= 0, \qquad\quad \forall \omega \in \mathbb{W}_S. \end{aligned} \tag{2.58}$$

We introduce two finite-dimensional subspaces $\mathbb{V}_{S,h} \subset \mathbb{V}$ and $\mathbb{W}_{S,h} \subset \mathbb{W}$, and consider the discretized problem:

Find $(\mathbf{u}_h, p_h) \in \mathbb{V}_{S,h} \times \mathbb{W}_{S,h}$ such that

$$\begin{aligned} (\nabla \mathbf{u}_h, \nabla \mathbf{v}_h) - (p_h, \nabla \cdot \mathbf{v}_h) &= (\mathbf{f}, \mathbf{v}_h), \quad \forall \mathbf{v}_h \in \mathbb{V}_{S,h}, \\ (\nabla \cdot \mathbf{u}_h, \omega_h) &= 0, \qquad\quad \forall \omega_h \in \mathbb{W}_{S,h}. \end{aligned} \tag{2.59}$$

Now we construct a pair of discretized space $\mathbb{V}_{S,h} \times \mathbb{W}_{S,h}$ for this Stokes model problem in the next subsection.

2.4.5 Modified Bernardi-Raugel Element

Stable mixed finite element pairs for the Stokes problem have been widely and thoroughly studied on triangles and rectangles. One notable construction on rectangles was discussed in Bernardi and Raugel (1985) and Fortin (1981), where polynomial spaces are supplemented with bubble functions. Consider the standard type of Bernardi-Raugel (BR) element using the space of tensor product polynomials

$$(\mathbb{Q}_{1,1}(E))^2 \oplus \operatorname{span}\{\mathbf{b}_0, \mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\} \subset (\mathbb{Q}_{2,2}(E))^2, \tag{2.60}$$

where $\mathbf{b}_i$ is a quadratic bubble function defined for each edge e_i. A similar but more general type of element was discussed by Arbogast and Wheeler (2005).

We now modify the original BR element to extend it to a new finite element suited to quadrilaterals using the direct serendipity space as

$$\mathbb{V}_1^{\mathcal{BR}}(E) = (\mathcal{DS}_1(E))^2 \oplus \operatorname{span}\{\mathbf{b}_0, \mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\} \subset (\mathcal{DS}_2(E))^2, \tag{2.61}$$

where $\dim \mathbb{V}_1^{\mathcal{BR}}(E) = 12$. One natural choice for the supplemental bubble function $\mathbf{b}_i$ is the previously defined edge nodal basis function $\varphi_{e,i}^{(2)} \in \mathcal{DS}_2$ times the unit normal, i.e.,

$$\mathbf{b}_i(\mathbf{x}) = \varphi_{e,i}^{(2)}(\mathbf{x})\boldsymbol{\nu}_i, \tag{2.62}$$

which is quadratic when restricted to edge e_i according to (2.21). The quadratic bubble is guaranteed to be conforming since it evaluates to 0 on two vertices and 1 at the middle point of the edge due to normalization, which is uniquely determined by these three points. The integral of this quadratic bubble can be accurately computed by a three-point Gaussian quadrature rule.

We state the degrees of freedom of this type of element as follows.

Definition 2.5. For an element $E \in \mathcal{T}_h$, $\mathbf{u}_h$ is uniquely defined by the following 12 degrees of freedom:

(i) for corner point v_i and Cartesian direction unit normal vectors $\{\mathbf{i}, \mathbf{j}\}$,

$$DOF_{(v_i,\mathbf{i})}^{(\mathrm{i})}(\mathbf{u}_h) = \mathbf{u}_h(v_i) \cdot \mathbf{i}, \quad DOF_{(v_i,\mathbf{j})}^{(\mathrm{i})}(\mathbf{u}_h) = \mathbf{u}_h(v_i) \cdot \mathbf{j};$$

(ii) for edge e_i,

$$DOF_{e_i}^{(ii)} = \langle \mathbf{u}_h \cdot \boldsymbol{\nu}_i, 1 \rangle.$$

We give an explicit construction of the shape functions. We choose the nodal basis function on vertex v_i in two Cartesian directions as

$$N_{v_i,x} = \begin{bmatrix} \varphi_{v,i}^{(1)}(\mathbf{x}) \\ 0 \end{bmatrix}, \quad N_{v_i,y} = \begin{bmatrix} 0 \\ \varphi_{v,i}^{(1)}(\mathbf{x}) \end{bmatrix}, \tag{2.63}$$

where $\varphi_{v,i}^{(1)}$ is the vertex nodal shape function in $\mathcal{DS}_1(E)$. We choose the edge basis function on edge e_i as (2.62), i.e.,

$$N_{e,i} = \mathbf{b}_i(\mathbf{x}) = \varphi_{e,i}^{(2)}\boldsymbol{\nu}_i \tag{2.64}$$

Note that bubble functions are quadratic when restricted to edge e_i, while nodal basis functions are linear when restricted to edge e_i. Therefore the bubble functions cannot be in the same space as the nodal basis functions. We conclude that this element is well-defined, i.e., the degrees of freedom uniquely determine the function.

We pair it with the space

$$\mathbb{W}_0 = \{\omega \in L^2(\Omega); \quad \omega|_E \in P_0\}/\mathbb{R}, \tag{2.65}$$

which has piecewise constant values for each element E. We provide stability and approximation analysis for this space pair $\mathbb{V}_1^{\mathcal{BR}} \times \mathbb{W}_0$ in the next subsection.

2.4.6 Stability and Approximation Properties for $\mathbb{V}_1^{\mathcal{BR}} \times \mathbb{W}_0$

Before analyzing the stability and approximation properties for the new modified BR element, we state one useful lemma.

Lemma 2.6. *Let E be a nondengenate convex quadrilateral element that is shape regular as in Definition 2.1. For each integer $m \geq 0$ and real number $p \in [1, \infty]$ there exist positive constants C_1 and C_2, possibly depending on σ^* but otherwise independent of the geometry of E, such that the following upper bounds hold for all $v \in W_p^m(E)$:*

$$\begin{cases} |v|_{W_p^0(E)} \leq C_1 h_E^{2/p} |\tilde{v}|_{W_p^0(\tilde{E})}, \\ |v|_{W_p^1(E)} \leq C_2(\sigma^*, p) h^{2/p-1} |\tilde{v}|_{W_p^1(\tilde{E})}, \end{cases} \tag{2.66}$$

where $\tilde{E}$ represents the local reference quadrilateral element $\tilde{E}$ illustrated in Figure 2.1, and $\tilde{v}$ is the function defined on the local reference element $\tilde{E}$.

Proof. See Girault and Raviart (2011). □

We reuse the interpolation operator $\mathcal{I}_h^1$ defined in equation (2.29). Define the projection operator $\pi \in (H_0^1(\Omega))^2 \to \mathbb{V}_1^{\mathcal{BR}} \cap (H_0^1(\Omega))^2$ as a combination of local projection operators π_E as:

(i) For each vertex $v \in \partial E$

$$\pi_E \mathbf{v}(\mathbf{x}_v) = \mathcal{I}_h^1 \mathbf{v}(\mathbf{x}_v); \tag{2.67}$$

(ii) For each edge $e \subset \partial E$,

$$\int_e (\pi_E \mathbf{v} - \mathbf{v}) \cdot \boldsymbol{\nu} \, ds = 0. \tag{2.68}$$

Lemma 2.7. *For all* $\mathbf{v} \in (H^k(\Omega))^2 \cap (H_0^1(\Omega))^2, \quad k = 1, 2$, *the operator* π_E *defined by* (2.67)–(2.68) *satisfies*

$$(q_h, \nabla \cdot (\mathbf{v} - \pi_E \mathbf{v})) = 0, \quad \forall q_h \in \mathbb{W}_0 \tag{2.69}$$

Assuming shape regularity as defined in Definition 2.1, *the global projection operator* π *satisfies the error bound*

$$\|\mathbf{v} - \pi \mathbf{v}\|_{H^1(\Omega)} \leq C h^m \|\mathbf{v}\|_{H^{m+1}(\Omega)}, \quad \forall \mathbf{v} \in (H^{m+1}(\Omega))^2, \ m = 0, 1. \tag{2.70}$$

Proof. According to the divergence theorem, we have

$$\int_E \nabla \cdot (\mathbf{v} - \pi_E \mathbf{v}) \, d\mathbf{x} = \sum_{i=0}^{3} \int_{e_i} (\mathbf{v} - \pi_E \mathbf{v}) \cdot \boldsymbol{\nu} \, ds = 0. \tag{2.71}$$

Given that q_h is piecewise constant on the element E, it is clear that we have

$$(q_h, \nabla \cdot (\mathbf{v} - \pi_E \mathbf{v})) = 0, \quad \forall q_h \in \mathbb{W}_0. \tag{2.72}$$

From the definition of the degrees of freedom (2.67) and (2.68), we decompose the projection operator π_E as

$$\pi_E \mathbf{v} = \mathcal{I}_h^1 \mathbf{v} + \sum_{i=0}^{3} \alpha_i \mathbf{b}_i, \tag{2.73}$$

where the coefficient

$$\alpha_i = \frac{\int_{e_i} (\mathbf{v} - \mathcal{I}_h^1 \mathbf{v}) \cdot \boldsymbol{\nu}_i \, ds}{\int_{e_i} \mathbf{b}_i \cdot \boldsymbol{\nu}_i \, ds}. \tag{2.74}$$

The approximation property of the interpolation operator $\mathcal{I}_h^1$ has been discussed in Lemma 2.4,

$$\left\|\mathbf{v} - \mathcal{I}_h^1 \mathbf{v}\right\|_{H^m(E)} \leq h^{n-m} |\mathbf{v}|_{H^n(E^*)}, \quad m = 0, 1, \quad n = 1, 2. \tag{2.75}$$

where $E^* = \cup_{F\in\mathfrak{T}_h, F\cap E\neq\emptyset}$ defined in Lemma 2.3.

Lemma 2.6 indicates an estimation of the bubble function as

$$|\mathbf{b}_i|_{H^m(E)} \leq C(\sigma^*)h^{1-m}|\tilde{\mathbf{b}}_i|_{H^1(\tilde{E})}, \quad m=0,1. \tag{2.76}$$

Because of the shape regularity, the vertices of the local reference elements $\tilde{E}$ vary in a compact set. Moveover, as in Arbogast et al. (2022), the definition of $\tilde{\mathbf{b}}_i$ depends only on the vertices in a C^1 manner. Thus the maximum over the elements is finite, and so

$$|\tilde{\mathbf{b}}_i|_{H^1(\tilde{E})} \leq C(\sigma^*), \quad \forall E\in\mathfrak{T}_h. \tag{2.77}$$

Therefore

$$|\tilde{\mathbf{b}}_i|_{H^m(E)} \leq C(\sigma^*)h^{1-m}, \quad m=0,1. \tag{2.78}$$

We continue to estimate $|\alpha_i|$. Note

$$\int_{e,i}\mathbf{b}_i\cdot\boldsymbol{\nu}_i\,ds = \int_{e,i}\varphi_{e,i}^{(2)}\,ds = |e_i|\int_{\tilde{e}_i}\tilde{\varphi}_{e,i}^{(2)}\,d\tilde{s} = C_1|e_i|. \tag{2.79}$$

Then

$$\begin{aligned} |\alpha_i| = \left|\frac{\int_{e_i}(\mathbf{v}-\mathfrak{I}_h^1\mathbf{v})\cdot\boldsymbol{\nu}_i\,ds}{\int_{e_i}\mathbf{b}_i\cdot\boldsymbol{\nu}_i\,ds}\right| &\leq C\int_{\tilde{e},i}|\tilde{\mathbf{v}}-\tilde{\mathfrak{I}}_h^1\tilde{\mathbf{v}}|\,ds \\ &\leq C\left\|\tilde{\mathbf{v}}-\tilde{\mathfrak{I}}_h^1\mathbf{v}\right\|_{L^2(\tilde{e}_i)}. \end{aligned} \tag{2.80}$$

As the trace mapping is contimuous from $H^1(\tilde{E})$ into $L^2(\tilde{e})$, according to the Trace Theorem 2.1,

$$\begin{aligned} |\alpha_i| &\leq C\left\|\tilde{\mathbf{v}}-\tilde{\mathfrak{I}}_h^1\mathbf{v}\right\|_{H^1(\tilde{E})} \\ &\leq C\Big(\left\|\tilde{\mathbf{v}}-\tilde{\mathfrak{I}}_h^1\mathbf{v}\right\|_{L^2(E)} + h^{-1}\left|\tilde{\mathbf{v}}-\tilde{\mathfrak{I}}_h^1\mathbf{v}\right|_{H^1(E)}\Big). \end{aligned} \tag{2.81}$$

According to Lemma 2.4,

$$|\alpha_i| \leq Ch^{k-1}\left\|\mathbf{v}\right\|_{H^k(E^*)}, \quad k=1,2. \tag{2.82}$$

The overall estimate is then

$$|\alpha_i||\mathbf{b}_i|_{H^m(E)} \leq Ch^{k-m}\left\|\mathbf{v}\right\|_{H^k(E^*)}, \quad m=0,1, \quad k=1,2. \tag{2.83}$$

Combining with the interpolation error estimate result in equation (2.75), we get a local error estimate as

$$\|\mathbf{v} - \pi_E \mathbf{v}\|_{H^1(E)} \leq Ch \|\mathbf{v}\|_{H^2(E)} . \tag{2.84}$$

The global error estimate is obtained by combining local results together. □

In addition to the approximability lemma 2.4, we also need a stability result.

Lemma 2.8. *Assuming that* $\mathbf{v} \in (H^1(\Omega))^2$*, the linear operator* π *is bounded on* $(H^1(\Omega))^2$ *independently of* h*, i.e.,*

$$\|\pi \mathbf{v}\|_{H^1(\Omega)} \leq C \|\mathbf{v}\|_{H^1(\Omega)} . \tag{2.85}$$

Proof. According to the triangle inequality,

$$\|\pi_E \mathbf{v}\|_{H^1(E)} \leq \|\mathbf{v}\|_{H^1(E)} + \|\mathbf{v} - \pi_E \mathbf{v}\|_{H^1(E)} \leq (1 + C) \|\mathbf{v}\|_{H^1(E)} , \tag{2.86}$$

where C comes from Lemma 2.7. The global stability result can be obtained by combining local results. □

Theorem 2.9. *With the established approximability result stated in Lemma* 2.7*, and the stability result stated in Lemma* 2.8*, for a convex domain* Ω*, there exists a constant* $\gamma > 0$ *such that*

$$\inf_{q_h \in \mathbb{W}_0} \sup_{\mathbf{v}_h \in \mathbb{V}_1^{\mathcal{BR}}} \frac{(\nabla \cdot \mathbf{v}_h, q_h)}{\|\mathbf{v}_h\|_{H^1(\Omega)} \|q_h\|_{L^2(\Omega)}} \geq \gamma > 0. \tag{2.87}$$

Proof. Note that

$$\sup_{\mathbf{v}_h \in \mathbb{V}_1^{\mathcal{BR}}} \frac{(\nabla \cdot \mathbf{v}_h, q_h)}{\|\mathbf{v}_h\|_{H^1(\Omega)} \|q_h\|_{L^2(\Omega)}} \geq \frac{(\nabla \cdot \pi \mathbf{v}, q_h)}{\|\pi \mathbf{v}\|_{H^1(\Omega)} \|q_h\|_{L^2(\Omega)}}. \tag{2.88}$$

Recall that the inf-sup condition holds for the continuous space as

$$\inf_{q \in L^2(\Omega)} \sup_{\mathbf{v} \in (H^1(\Omega))^2} \frac{(\nabla \cdot \mathbf{v}, q)}{\|\mathbf{v}\|_{H^1(\Omega)} \|q\|_{L^2(\Omega)}} \geq \gamma > 0. \tag{2.89}$$

With the condition $(q_h, \nabla \cdot (\mathbf{v} - \pi_E \mathbf{v})) = 0$ proved in Lemma 2.7 we have

$$\frac{(\nabla \cdot \pi \mathbf{v}, q_h)}{\|\pi \mathbf{v}\|_{H^1(\Omega)} \|q_h\|_{L^2(\Omega)}} = \frac{(\nabla \cdot \mathbf{v}, q_h)}{\|\pi \mathbf{v}\|_{H^1(\Omega)} \|q_h\|_{L^2(\Omega)}} \geq \frac{\gamma}{C} > 0, \tag{2.90}$$

where C here is the constant in (2.85). $\square$

We conclude that our space $\mathbb{V}_1^{\mathcal{BR}} \times \mathbb{W}_0$ forms a inf-sup stable pair with first order overall accuracy.

2.4.7 Numerical Test

We test our modified Bernardi-Raugel elements on the incompressible isotropic elasticity problem but with a non-homogenous Dirichlet boundary condition, i.e., on the problem:

Find the displacement $\mathbf{u}$ and pressure p such that

$$\begin{aligned} -\Delta \mathbf{u} + \nabla p &= \mathbf{f} \quad \text{in } \Omega, \\ \nabla \cdot \mathbf{u} &= 0 \quad \text{in } \Omega, \\ \mathbf{u} &= \mathbf{g} \quad \text{on } \partial\Omega. \end{aligned} \tag{2.91}$$

We assume that $\mathbf{g} \in (H^{1/2}(\partial\Omega))^2$ and understand $\tilde{\mathbf{g}} \in (H^1(\Omega))^2$ as a continuous extension from the boundary into the domain. We define a simple choice of energy spaces

$$\begin{aligned} \mathbb{V} &= \{\mathbf{v} \in (H^1(\Omega))^2 : \mathbf{v} = \mathbf{0} \quad \text{on } \partial\Omega\}, \\ \mathbb{U} &= \{\mathbf{u} \in (H^1(\Omega))^2 : \mathbf{u} = \mathbf{g} \quad \text{on } \partial\Omega\} = \tilde{\mathbf{g}} + \mathbb{V}, \\ \mathbb{W} &= \{p \in L^2(\Omega)\}/\mathbb{R}. \end{aligned} \tag{2.92}$$

We write the problem in weak form as:

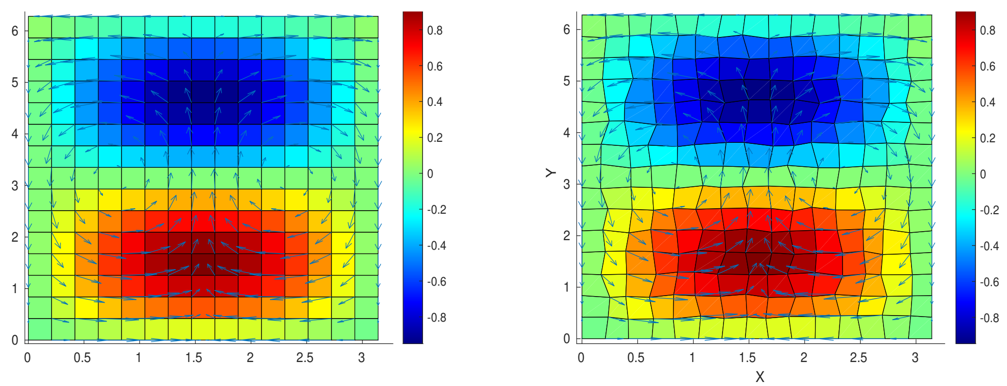
Find $\mathbf{u} \in \mathbb{U}$ and $p \in L^2(\Omega)$ such that

$$\begin{aligned} (\nabla \mathbf{u}, \nabla \mathbf{v}) - (p, \nabla \cdot \mathbf{v}) &= (\mathbf{f}, \mathbf{v}), \quad \forall \mathbf{v} \in \mathbb{V}, \\ (\nabla \cdot \mathbf{u}, \omega) &= 0, \qquad \forall \omega \in \mathbb{W}. \end{aligned} \tag{2.93}$$

We consider the true solution of test problem (2.93) defined on a square region $\Omega = [0, \pi] \times [0, 2\pi]$. We manufacture a displacement and pressure as

$$\mathbf{u} = \begin{bmatrix} \cos(x)\sin(y) \\ -\sin(x)\cos(y) \end{bmatrix}, \quad p = \sin(x)\sin(y), \tag{2.94}$$

which has 0 normal flux on the boundary. We deal with the constant kernal by enforcing $\int_\Omega p \, d\mathbf{x} = 0$ and solve the linear system directly. The numerical results are computed on rectangular and logically rectangular quadrilateral meshes as depicted in Figures 2.10a and 2.10b, respectively. We calculate the L^2 norm of the residual (i.e., the difference) of the true and approximate pressure, velocity, and derivative of velocity.



(a) Rectangular Grids. Shown are pressure (color) and velocity (arrows).

(b) Distorted Grids. Shown are pressure (color) and velocity (arrows).

Figure 2.10: Numerical results on rectangular and quadrilateral mesh.

In Table 2.1 and Table 2.2 we display the result of a calculation of convergence on $n \times n$ rectangular and distorted grids, respectively, for $n = 8, 16, 32, 64$. A second order convergence rate is achieved by velocity. Pressure achieves some superconvergence due to the use of the mid point integration rule. We see second order convergence on the rectangular grid, and some rate better than the first on distorted grid. A first order convergence rate is achieved by derivative of velocity. These meet our expectations for optimal convergence order.

Grid size	8×8	16×16	32×32	64×64
$L^2(\mathrm{res}_u)$	4.502E−1	1.052E−1	2.513E−2	6.130E−3
$L^2(\mathrm{res}_{du})$	1.039E+0	4.463E−1	2.093E−1	1.019E−1
$L^2(\mathrm{res}_p)$	2.632E+0	5.845E−1	1.374E−1	3.330E−2

Table 2.1: Rectangular Grids. L^2-error in the residual of the solution u (res_u), the pressure p (res_p), and the of derivatives of the solution Du (res_{du}).

Grid size	8×8	16×16	32×32	64×64
$L^2(\mathrm{res}_u)$	4.603E−1	1.074E−2	2.567E−2	6.266E−3
$L^2(\mathrm{res}_{du})$	1.070E+0	4.670E−1	2.207E−1	1.078E−1
$L^2(\mathrm{res}_p)$	2.685E+0	5.981E−1	1.525E−1	4.580E−2

Table 2.2: Distorted Grids. L^2-error in the residual of the solution u (res_u), the pressure p (res_p), and the of derivatives of the solution Du (res_{du}).

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